[ADS5034-01] Computer Vision

13. Detection & Segmentation

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Outline

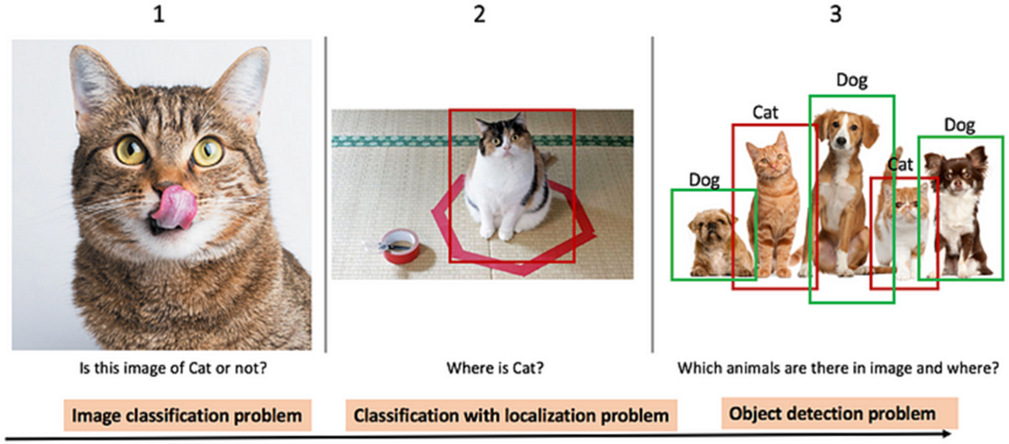
- Detection & Segmentation
- Object Detection
  - RCNN (Fast/Faster)
  - SSD / YOLO
- Segmentation
  - Encoder-Decoder
  - Mask-RCNN
- Data Set / Evaluation

Detection & Segmentation

- Overview -

Categorization of computer vision tasks

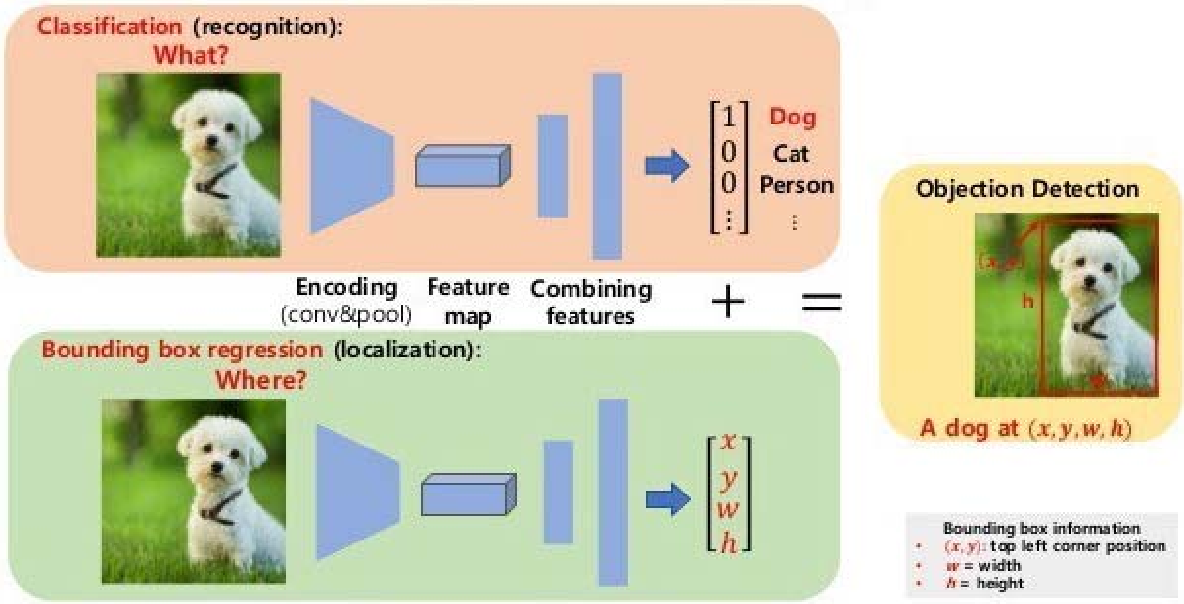
- **Image Classification: What's** in the image? This is the most basic task. The model assigns a single label to the entire image, like "cat" or "landscape."
- **Object Detection: What objects** are in the image, and **where** are they? This goes beyond classification. The model identifies specific objects (cats, dogs, people) and draws bounding boxes around them to indicate ***their location***.
- **Image Segmentation:** What are the **exact shapes of the objects** in the image? This provides the most detail. The model assigns a label to each individual pixel in the image, creating a kind of digital mask that ***outlines the shape of each object***.


1
2
3
Dog
Cat
Dog
Cat
Dog
Dog
Is this image of Cat or not?
Where is Cat?
Which animals are there in image and where?
Image classification problem
Classification with localization problem
Object detection problem

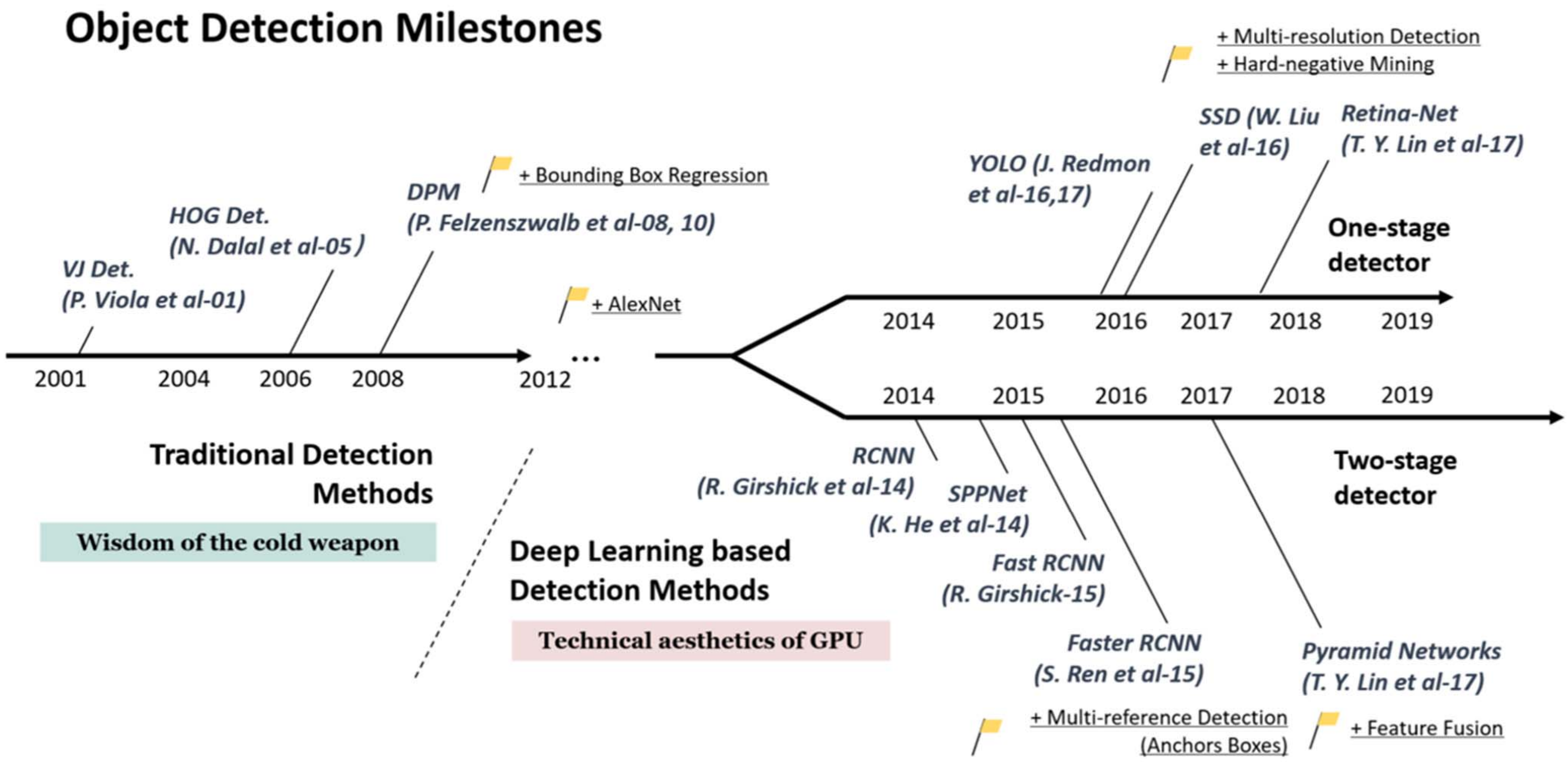
Detection

- Object detection is a computer vision solution that identifies **what** is in an image and **where** the detected objects are in the image.

Object Detection : Classification + Localization(Regression)

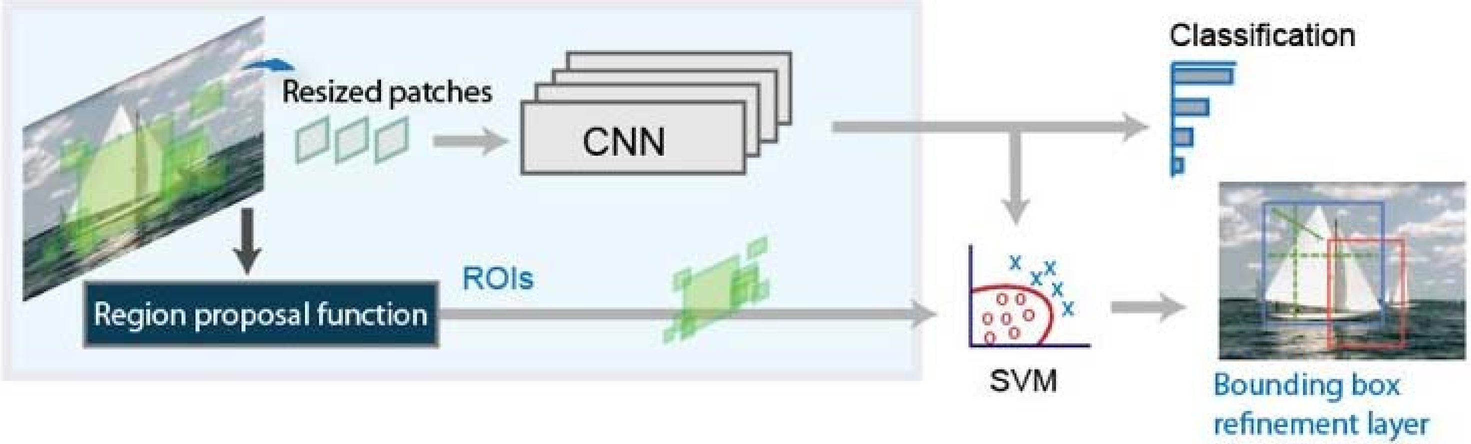

Classification (recognition):
What?
$\begin{bmatrix}1\\0\\0\\\vdots\end{bmatrix}$
Dog
Cat
Person
⋮
Encoding
(conv&pool)
Feature
map
Combining
features
+
=
Bounding box regression (localization):
Where?
$\begin{bmatrix}x\\y\\w\\h\end{bmatrix}$
Objection Detection
(x, y)
h
A dog at (x, y, w, h)
Bounding box information
• (x, y): top left corner position
• w = width
• h = height

Milestones in state-of-the-art Object Detection


Object Detection Milestones
VJ Det.
(P. Viola et al-01)
HOG Det.
(N. Dalal et al-05)
DPM
(P. Felzenszwalb et al-08, 10)
+ Bounding Box Regression
+ AlexNet
2001
2004
2006
2008
2012
...
Traditional Detection Methods
Wisdom of the cold weapon
Deep Learning based Detection Methods
Technical aesthetics of GPU
+ Multi-resolution Detection
+ Hard-negative Mining
YOLO (J. Redmon et al-16,17)
SSD (W. Liu et al-16)
Retina-Net
(T. Y. Lin et al-17)
One-stage detector
2014
2015
2016
2017
2018
2019
2014
2015
2016
2017
2018
2019
Two-stage detector
RCNN
(R. Girshick et al-14)
SPPNet
(K. He et al-14)
Fast RCNN
(R. Girshick-15)
Faster RCNN
(S. Ren et al-15)
Pyramid Networks
(T. Y. Lin et al-17)
+ Multi-reference Detection
(Anchors Boxes)
+ Feature Fusion

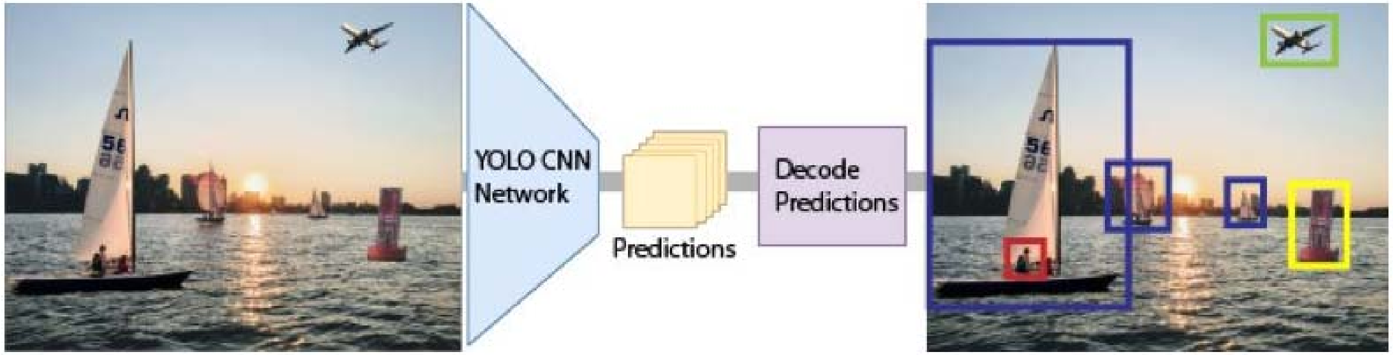
One-stage vs. two-stage object detectors

- **Two-stage detectors**: The approximate object regions are proposed using deep features before these features are used for the classification as well as bounding box regression for the object candidate.
  - The two-stage architecture involves (1) object region proposal with conventional Computer Vision methods or deep networks, followed by (2) object classification based on features extracted from the proposed region with bounding-box regression.
  - Various two-stage detectors include region convolutional neural network (RCNN), with evolutions Faster R-CNN, Mask R-CNN or granulated RCNN (G-RCNN).


Resized patches
CNN
Classification
Region proposal function
ROIs
SVM
Bounding box refinement layer

One-stage vs. two-stage object detectors

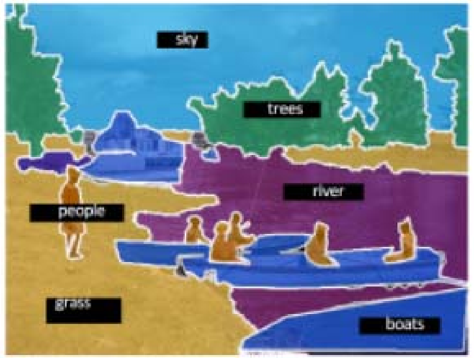
- **One-stage detectors**: It predicts bounding boxes over the images without the region proposal step. This process consumes less time and can therefore be used in real-time applications.
  - One-stage object detectors prioritize inference speed and are super fast but not as good at recognizing irregularly shaped objects or a group of small objects.
  - The most popular one-stage detectors include the YOLO-series, SSD, and RetinaNet.



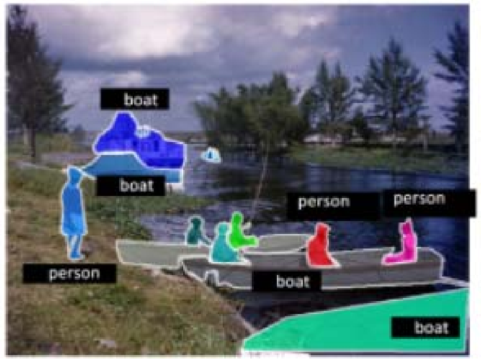
Segmentation

- **Goal:** Simplify and analyze images by separating them into different segments.

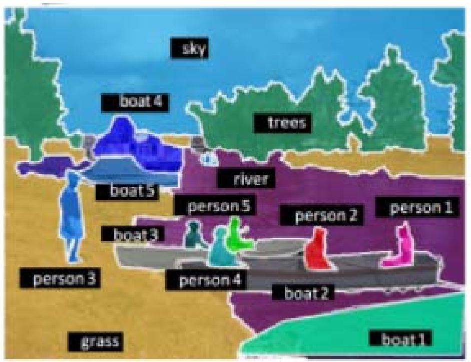
  This makes it easier for computers to understand the content of the image
  - Semantic Segmentation
  - Instance Segmentation
  - Panoptic Segmentation


sky
trees
river
people
grass
boats

Semantic Segmentation


boat
boat
person
person
person
boat
boat

Instance Segmentation

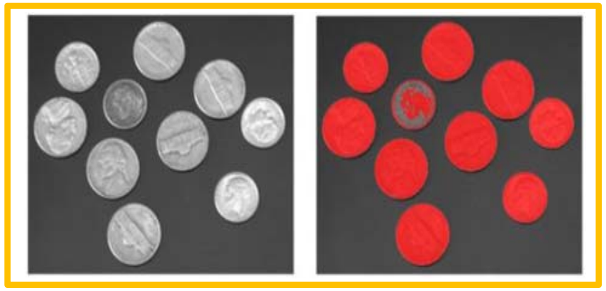

sky
boat 4
trees
river
boat 5
person 5
person 2
person 1
boat 3
person 3
person 4
boat 2
grass
boat 1

Panoptic Segmentation

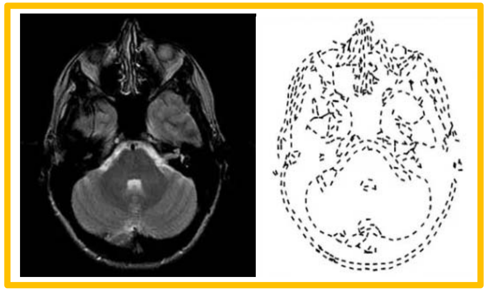
Segmentation

Methods (1)

- **Boundary-Based Segmentation:** Detects edges separating regions, aiding in object contour extraction.
- **Region-Based Segmentation:** Groups pixels based on similarity, using criteria like color or texture.
- **Clustering-Based Segmentation:** Utilizes clustering algorithms to group pixels with similar features.



Boundary-Based Segmentation

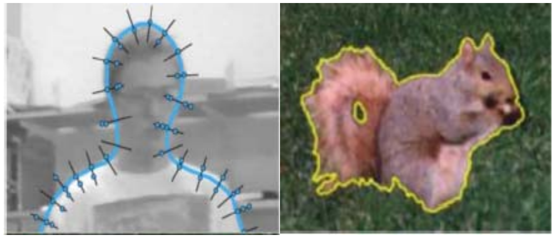


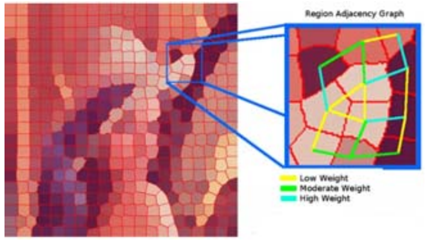
Region-Based Segmentation

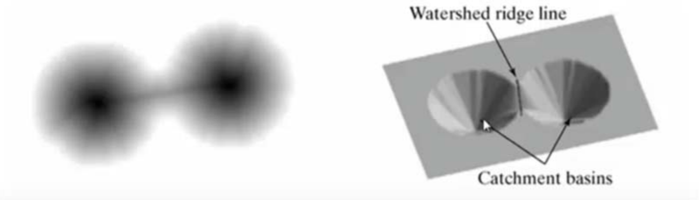
Clustering-Based Segmentation

Segmentation

- **Methods (2)**
  - **Active Contour Models (Snakes):** Flexible curves tracing object boundaries under various forces.
  - **Deep Learning-Based Segmentation:** Employs CNNs for accurate segmentation, with architectures like U-Net and Mask R-CNN.
  - **Graph-Based Segmentation:** Treats images as graphs, dividing them into segments using pixel connections.
  - **Watershed Segmentation:** Models pixel intensities as basins, useful for objects with clear intensity differences.


Active Contour Modes


Region Adjacency Graph
Low Weight
Moderate Weight
High Weight
Graph-Based Segmentation


Watershed ridge line
Catchment basins
Watershed Segmentation

Object Detection

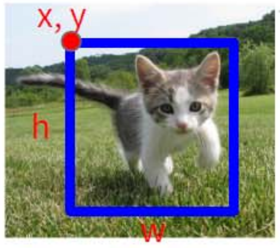
(R-CNN / Fast-RCNN / Faster-RCNN / YOLO)

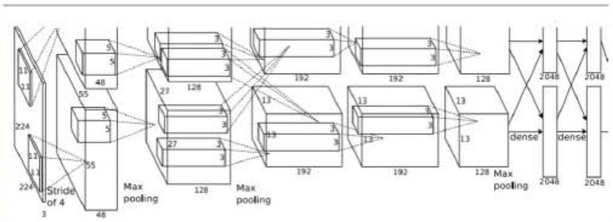
R-CNN (https://arxiv.org/abs/1311.2524)
Fast R-CNN (https://arxiv.org/abs/1504.08083)
Faster R-CNN (https://arxiv.org/abs/1506.01497)

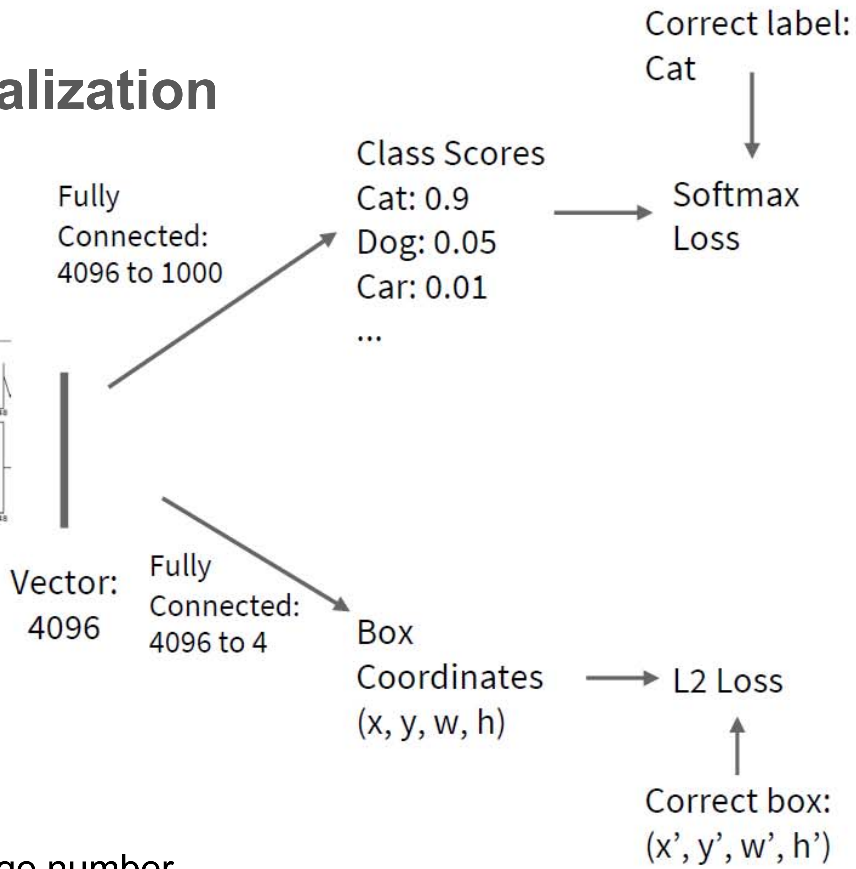
Object Detection

- Object Detection : **Classification + Localization**

Single Object




Stride of 4
Max pooling
Max pooling
Max pooling
dense
dense



Correct label:
Cat

Fully
Connected:
4096 to 1000

Class Scores
Cat: 0.9
Dog: 0.05
Car: 0.01
...

Softmax
Loss

Vector:
4096

Fully
Connected:
4096 to 4

Box
Coordinates
(x, y, w, h)

L2 Loss

Correct box:
(x', y', w', h')

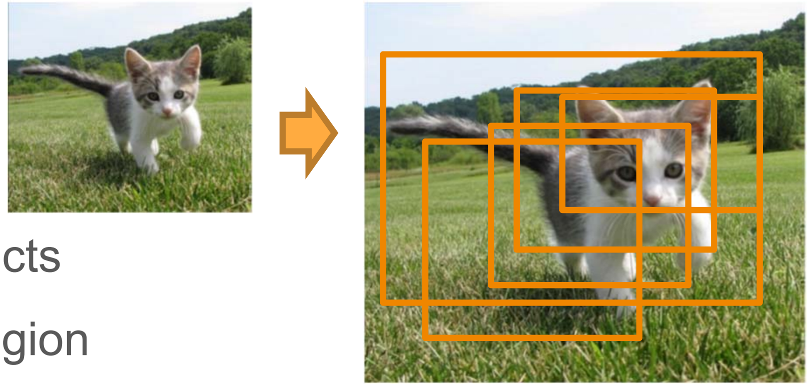
Multiple Objects

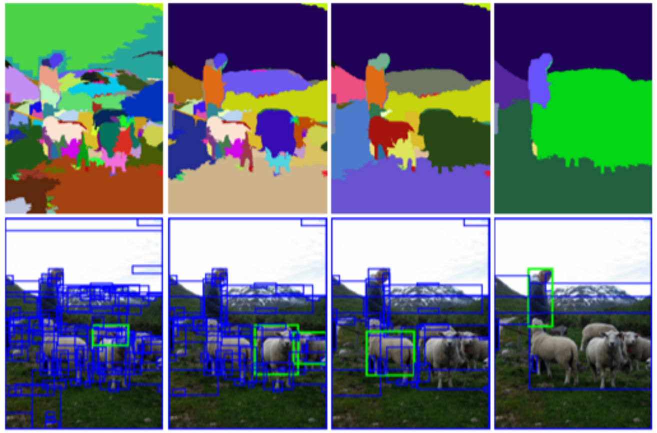
regression problem!

Need to apply CNN to huge number of locations, scales, and aspect ratios, very computationally expensive

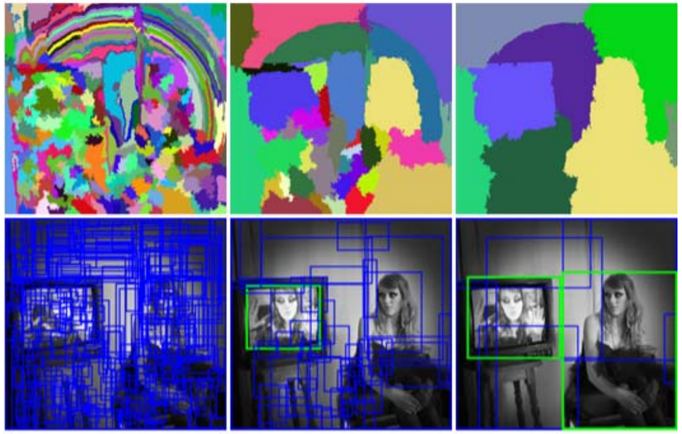
Region Proposals

- **Selective Search**
  - Find “blobby” image regions that are likely to contain objects
  - Relatively fast to run; e.g. Selective Search gives 2000 region proposals in a few seconds on CPU





Many objects at different scales

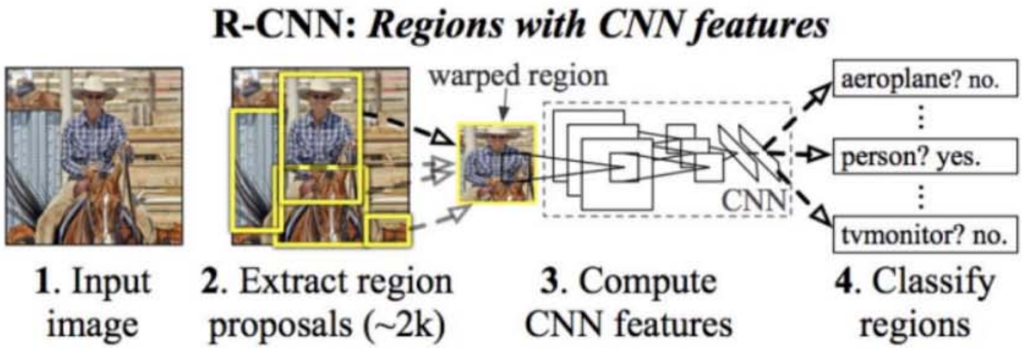


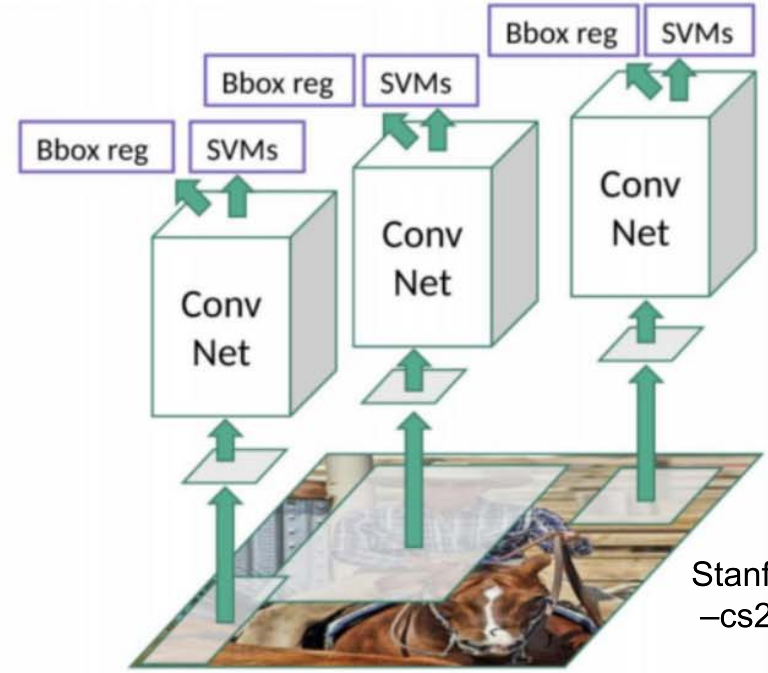
Special objects at different scales
as the girl is contained by the tv

Alexe et al, ”Measuring the objectness of image windows”, TPAMI 2012
Uijlings et al, ”Selective Search for Object Recognition”, IJCV 2013

R-CNN (Regions with CNN)

- Object detection model that uses high-capacity CNNs to bottom-up region
- It uses **selective search** to identify a number of bounding-box object region candidates (“regions of interest”), and then extracts features from each region independently for classification.


R-CNN: *Regions with CNN features*
warped region
aeroplane? no.
person? yes.
tvmonitor? no.
CNN
1. Input image
2. Extract region proposals (~2k)
3. Compute CNN features
4. Classify regions


Bbox reg
SVMs
Bbox reg
SVMs
Bbox reg
SVMs
Conv Net
Conv Net
Conv Net

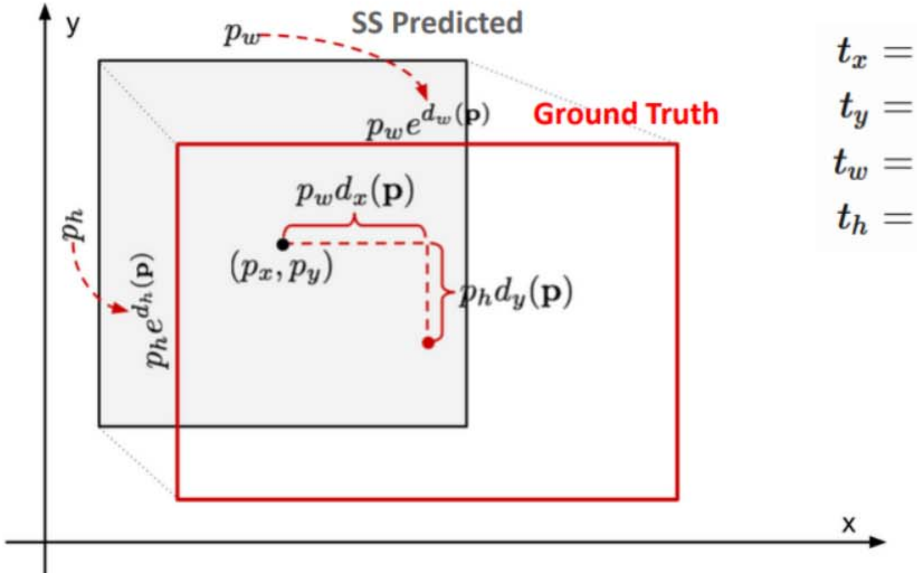
Stanford lecture
–cs231n

R-CNN (Regions with CNN)

- **Bounding Box Regression**

Adjusted Prediction

$$\hat{g}_x = p_w d_x(\mathbf{p}) + p_x$$
$$\hat{g}_y = p_h d_y(\mathbf{p}) + p_y$$
$$\hat{g}_w = p_w \exp(d_w(\mathbf{p}))$$
$$\hat{g}_h = p_h \exp(d_h(\mathbf{p}))$$


y
p_w
SS Predicted
$p_w e^{d_w(\mathbf{p})}$
Ground Truth
p_h
$p_h e^{d_h(\mathbf{p})}$
$p_w d_x(\mathbf{p})$
(p_x, p_y)
$p_h d_y(\mathbf{p})$
x

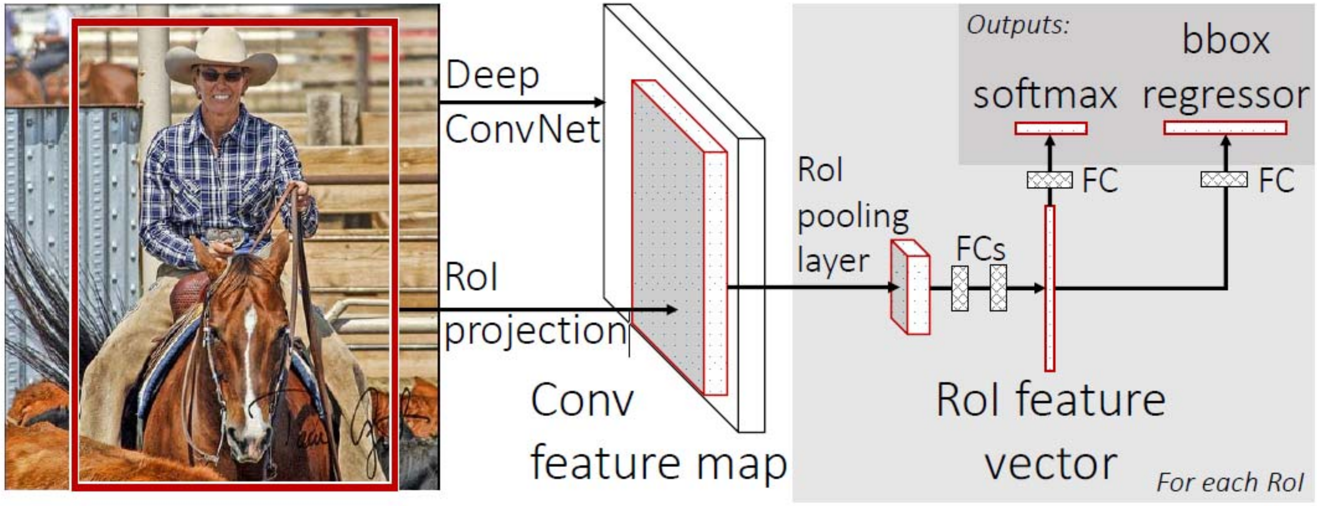
Target

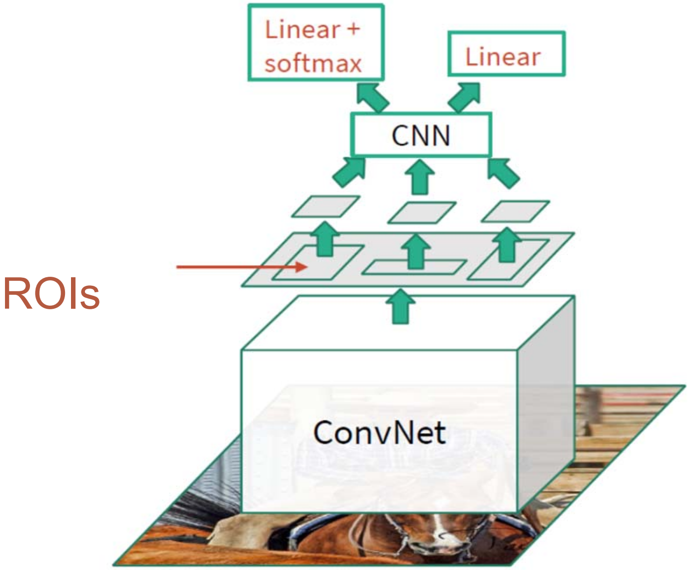
$$t_x = (g_x - p_x)/p_w$$
$$t_y = (g_y - p_y)/p_h$$
$$t_w = \log(g_w/p_w)$$
$$t_h = \log(g_h/p_h)$$

$$\mathcal{L}_{\text{reg}} = \sum_{i \in \{x,y,w,h\}} (t_i - d_i(\mathbf{p}))^2 + \lambda \|\mathbf{w}\|^2$$

Fast R-CNN

- Fast R-CNN is an improvement over R-CNN that addresses the computational inefficiency of the original method.
- It achieves this by using a single CNN to process the entire image and generate the ROIs rather than running the CNN on each ROI individually.

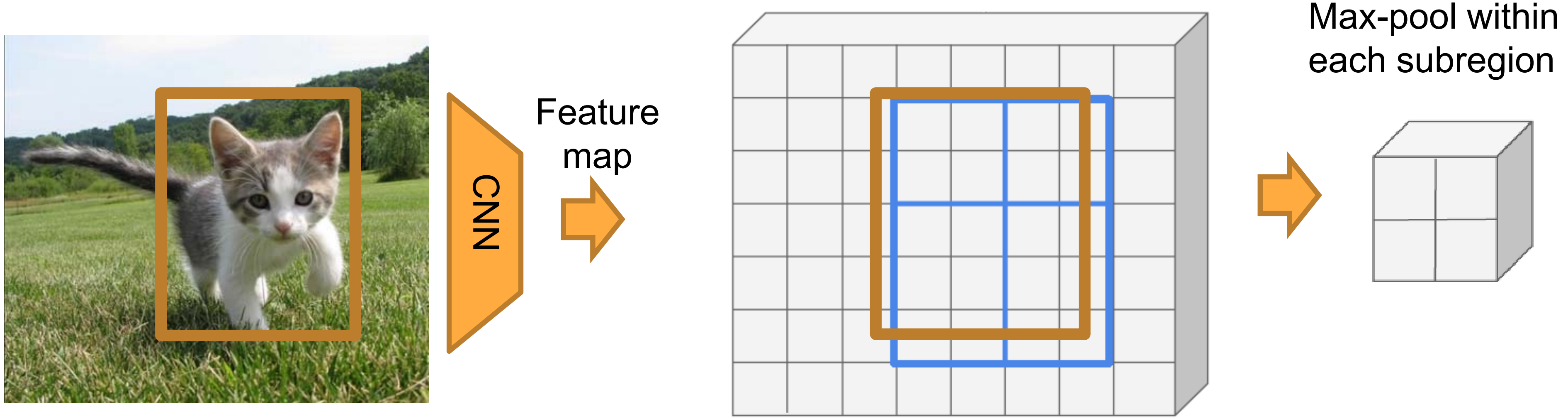

Deep
ConvNet
RoI
projection
Conv
feature map
RoI
pooling
layer
FCs
Outputs:
softmax
bbox
regressor
FC
FC
RoI feature
vector
For each RoI


Linear +
softmax
Linear
CNN
ROIs
ConvNet

Stanford lecture –cs231n

Fast R-CNN

- **RoI Pool (Cropping Features)**



(1) "Snap" to grid cells
(2) Divide into 2x2 grid of (roughly) equal subregions

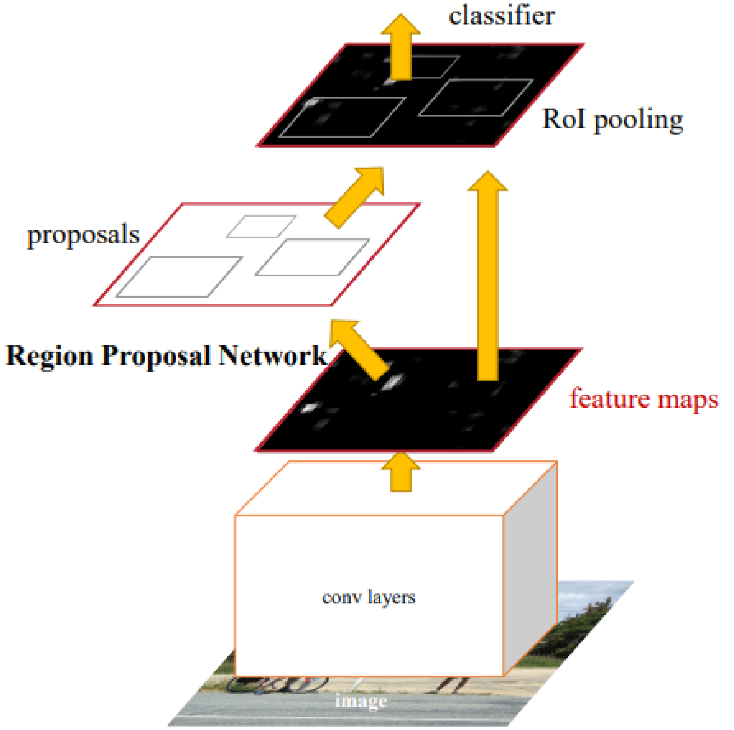
Max-pool within each subregion

Region features always the same size even if input regions have different sizes!

Problem: Region features slightly misaligned

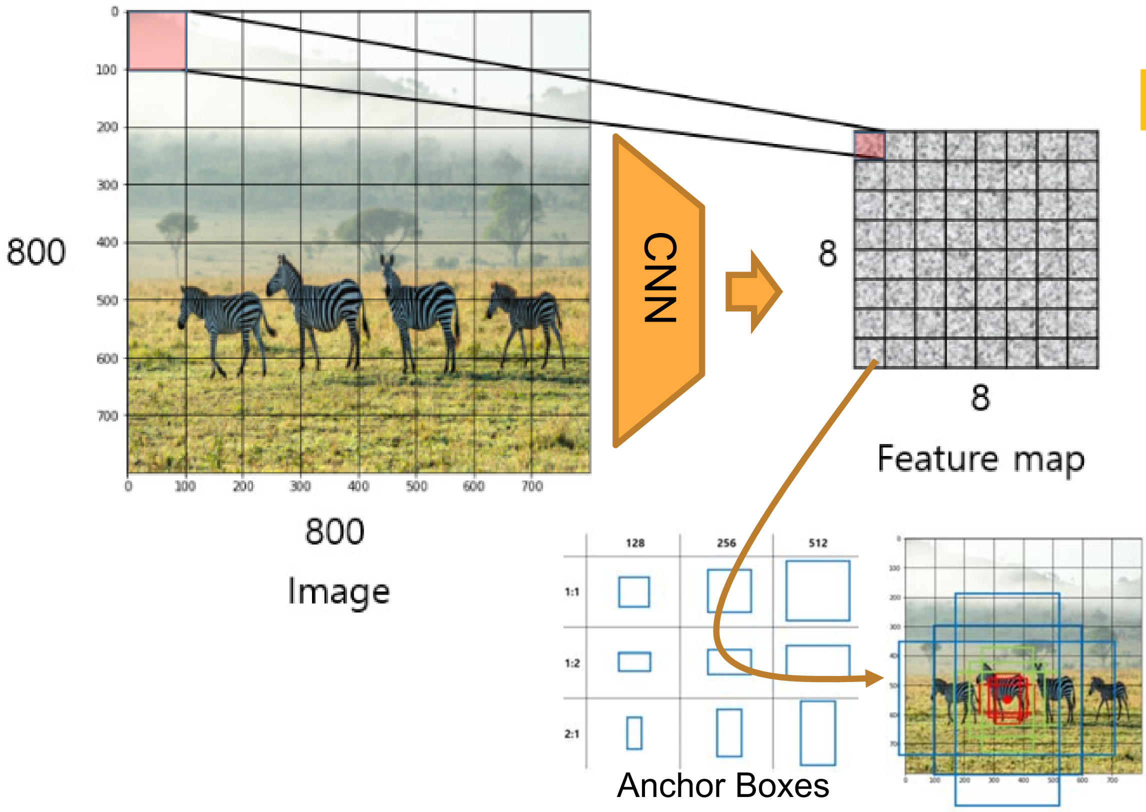
Faster R-CNN

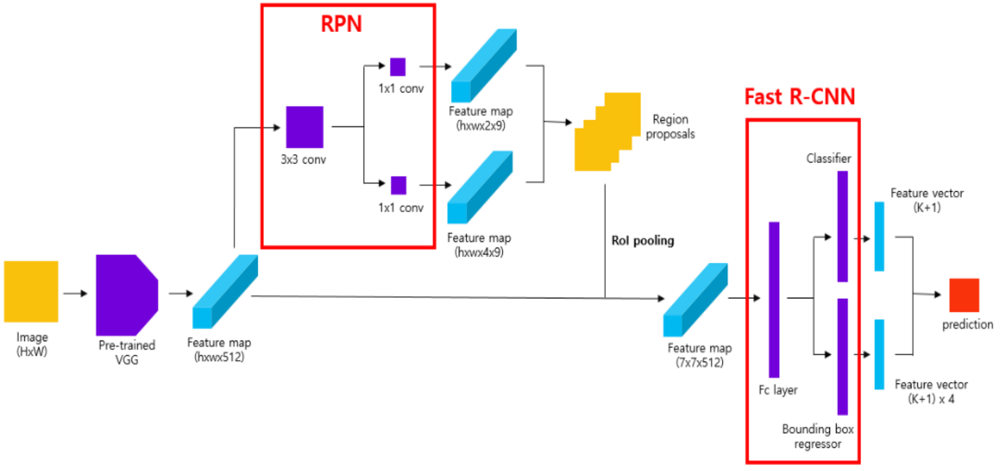
- This is later solved by the Faster R-CNN, which builds a region-proposal network that can generate region proposals that are fed to the detection model (Fast R-CNN) to inspect for objects.
- Faster R-CNN is a Two-stage object detector
  - First stage: Run once per image
    - Backbone network
    - **Region proposal network**
  - Second stage: Run once per region
    - Crop features: RoI pool / align
    - Predict object class
    - Prediction bbox offset

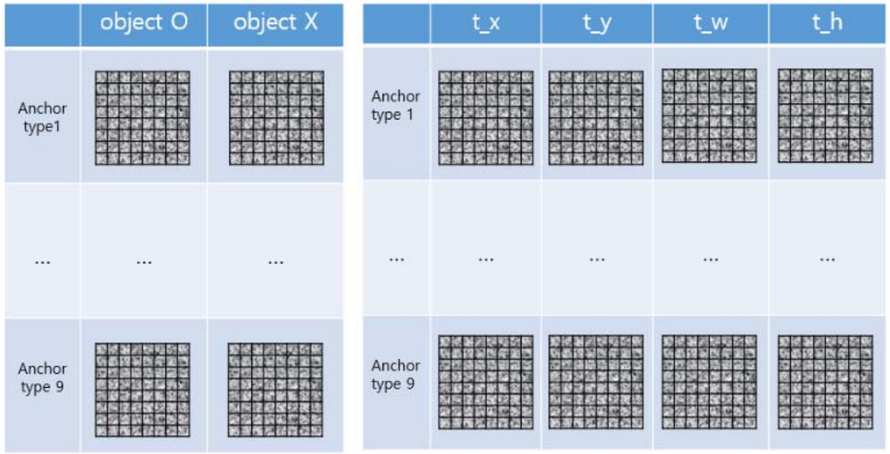

classifier
RoI pooling
proposals
Region Proposal Network
feature maps
conv layers
image

Faster R-CNN

- **Region Proposal Network**


0
100
200
300
400
500
600
700
800
800
Image
CNN
8
8
Feature map
128
256
512
1:1
1:2
2:1
Anchor Boxes


RPN
3x3 conv
1x1 conv
1x1 conv
Feature map (hxwx2x9)
Feature map (hxwx4x9)
Region proposals
RoI pooling
Fast R-CNN
Classifier
Feature vector (K+1)
prediction
Feature vector (K+1) x 4
Fc layer
Bounding box regressor
Image (HxW)
Pre-trained VGG
Feature map (hxwx512)
Feature map (7x7x512)

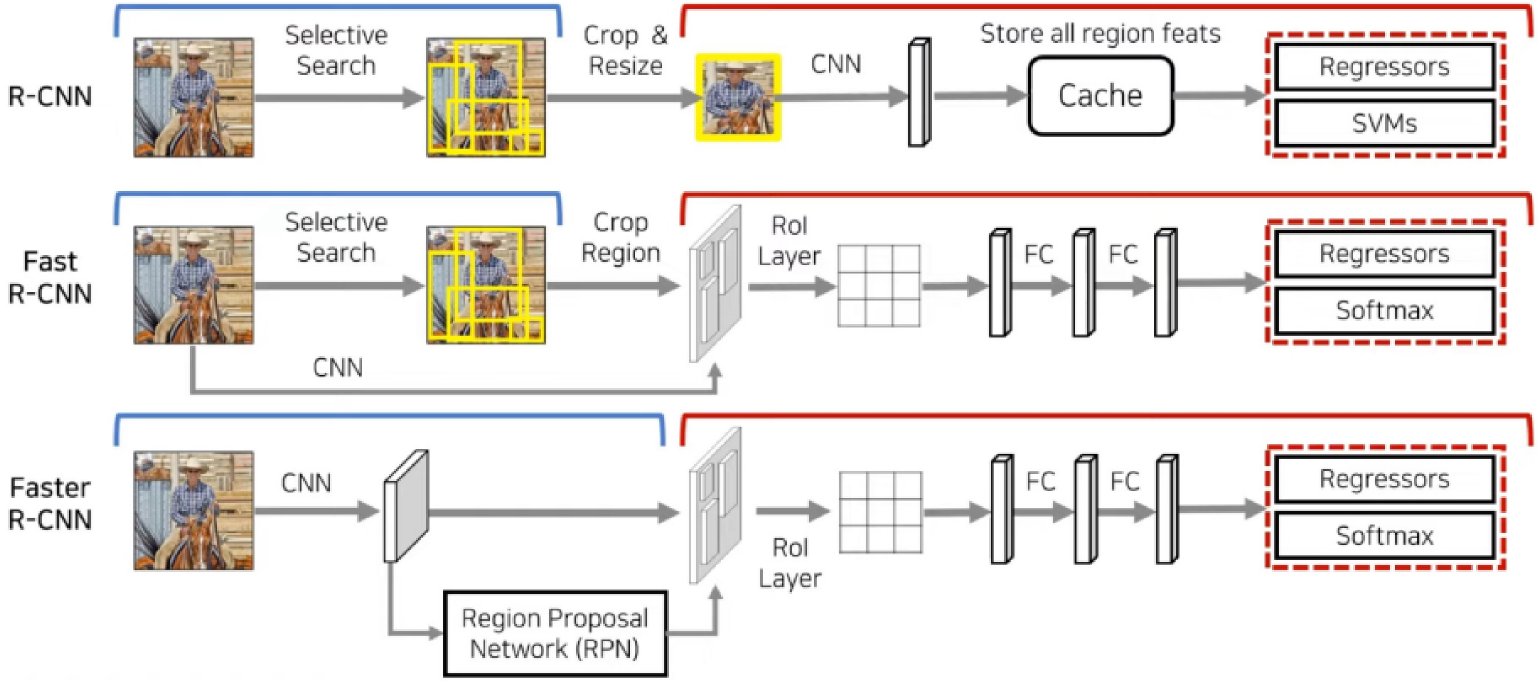


	object O	object X
Anchor type1		
…	…	…
Anchor type 9		

	t_x	t_y	t_w	t_h
Anchor type 1				
…	…	…	…	…
Anchor type 9				

Faster R-CNN

- **Region Proposal Network**

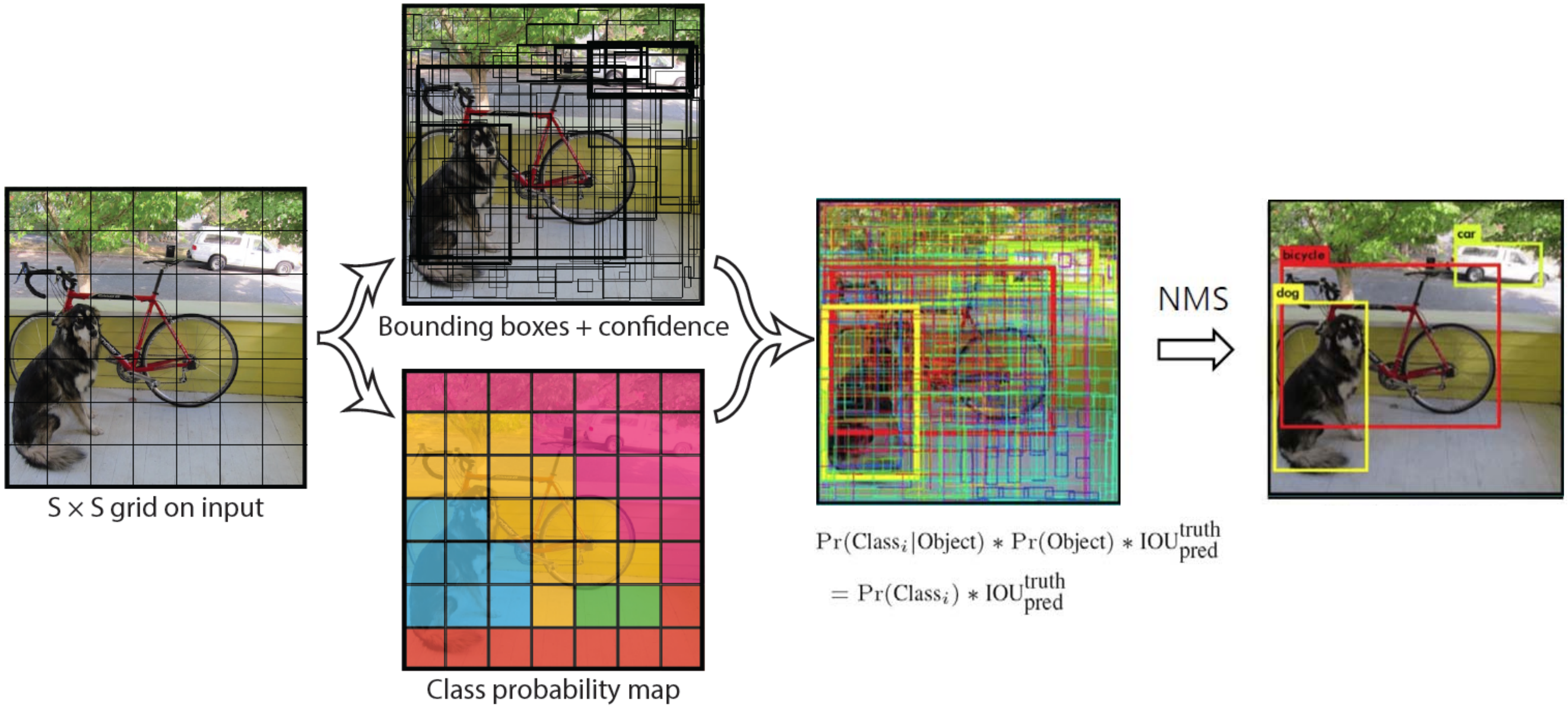

R-CNN
Selective Search
Crop & Resize
CNN
Store all region feats
Cache
Regressors
SVMs
Fast R-CNN
Selective Search
Crop Region
RoI Layer
FC
FC
Regressors
Softmax
CNN
Faster R-CNN
CNN
RoI Layer
FC
FC
Regressors
Softmax
Region Proposal Network (RPN)

Test time per image	Speed-up	mAP
50 sec.	1x	66.0%
2 sec.	25x	66.9%
0.2 sec.	250x	66.9%

https://www.arxiv-vanity.com/papers/1908.03673

YOLO (You Only Look Once)

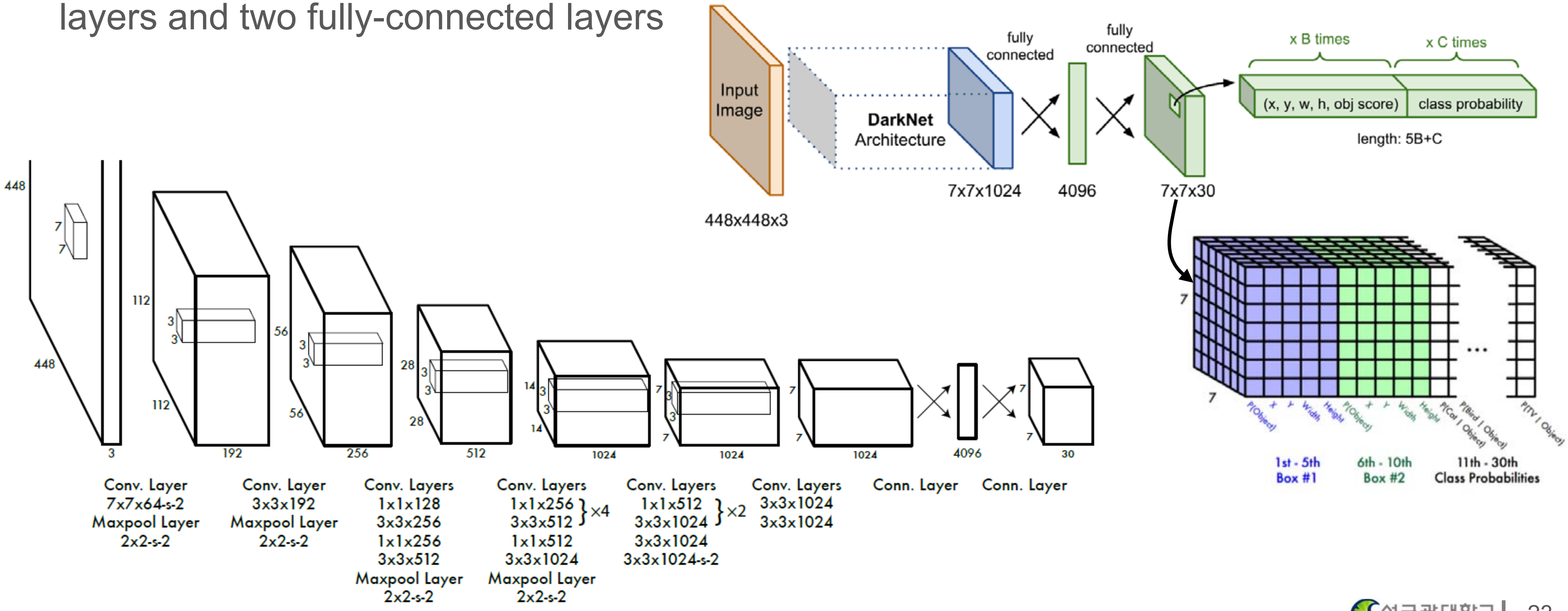
- Model


S × S grid on input
Bounding boxes + confidence
Class probability map
NMS
bicycle
car
dog

$$\Pr(\text{Class}_i|\text{Object}) * \Pr(\text{Object}) * \text{IOU}_{\text{pred}}^{\text{truth}} = \Pr(\text{Class}_i) * \text{IOU}_{\text{pred}}^{\text{truth}}$$

YOLO (You Only Look Once)

- YOLOv1 architecture is inspired by GoogleNet's architecture, and has 24 convolutional layers and two fully-connected layers


Input Image
448x448x3
DarkNet Architecture
fully connected
fully connected
7x7x1024
4096
7x7x30
x B times
x C times
(x, y, w, h, obj score)
class probability
length: 5B+C
448
7
7
448
3
112
3
3
112
192
56
3
3
56
256
28
3
3
28
512
14
3
3
14
1024
7
3
3
7
1024
7
7
1024
4096
7
7
30
Conv. Layer
7x7x64-s-2
Maxpool Layer
2x2-s-2
Conv. Layer
3x3x192
Maxpool Layer
2x2-s-2
Conv. Layers
1x1x128
3x3x256
1x1x256
3x3x512
Maxpool Layer
2x2-s-2
Conv. Layers
1x1x256
3x3x512 }×4
1x1x512
3x3x1024
Maxpool Layer
2x2-s-2
Conv. Layers
1x1x512
3x3x1024 }×2
3x3x1024
3x3x1024-s-2
Conv. Layers
3x3x1024
3x3x1024
Conn. Layer
Conn. Layer
7
7
P(Object)
X
Y
Width
Height
P(Object)
X
Y
Width
Height
P(Car | Object)
P(Bird | Object)
P(TV | Object)
...
1st - 5th
Box #1
6th - 10th
Box #2
11th - 30th
Class Probabilities

Redmon et al. "You only look once: unified, real-time object detection(2015)."

YOLO (You Only Look Once)

- Loss function

$$\lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 \right]$$

$$+ \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left[\left(\sqrt{w_i} - \sqrt{\hat{w}_i} \right)^2 + \left(\sqrt{h_i} - \sqrt{\hat{h}_i} \right)^2 \right]$$

Localization Loss

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{obj}} \left(C_i - \hat{C}_i \right)^2$$

$$+ \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^{B} \mathbb{1}_{ij}^{\text{noobj}} \left(C_i - \hat{C}_i \right)^2$$

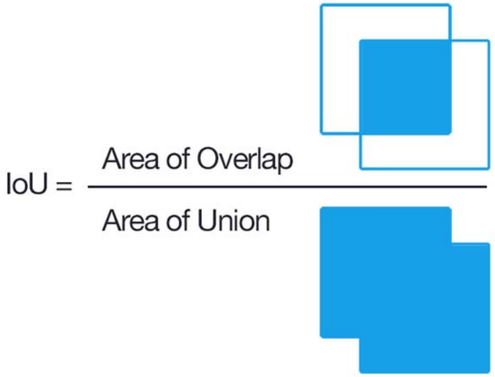
Confidence Loss

$$+ \sum_{i=0}^{S^2} \mathbb{1}_{i}^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2$$

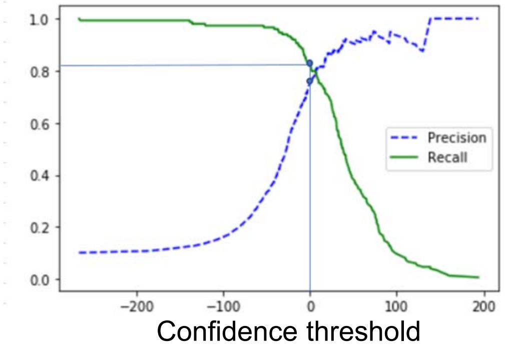
Classification Loss

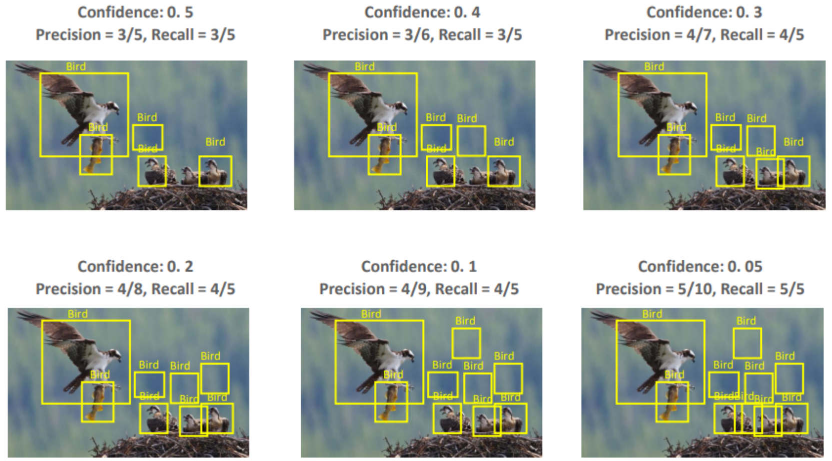
YOLO (You Only Look Once)

- IoU and Confidence score



$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$





Confidence: 0. 5
Precision = 3/5, Recall = 3/5

Confidence: 0. 4
Precision = 3/6, Recall = 3/5

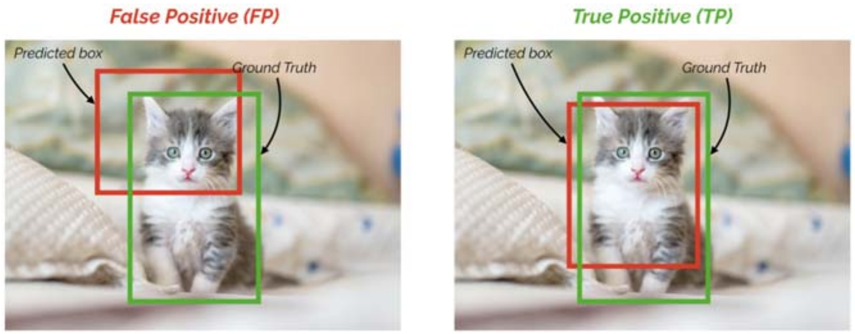
Confidence: 0. 3
Precision = 4/7, Recall = 4/5

Confidence: 0. 2
Precision = 4/8, Recall = 4/5

Confidence: 0. 1
Precision = 4/9, Recall = 4/5

Confidence: 0. 05
Precision = 5/10, Recall = 5/5

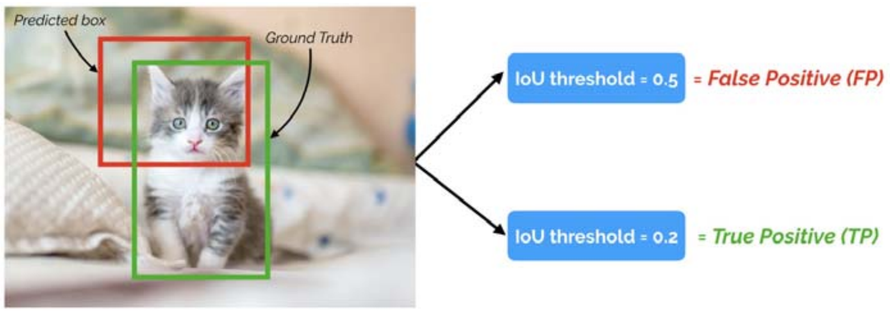
IoU threshold = 0.5


False Positive (FP)
True Positive (TP)
Predicted box
Ground Truth
Predicted box
Ground Truth

IoU = ~0.3

IoU = ~0.7

IoU Prediction = ~0.3


Predicted box
Ground Truth

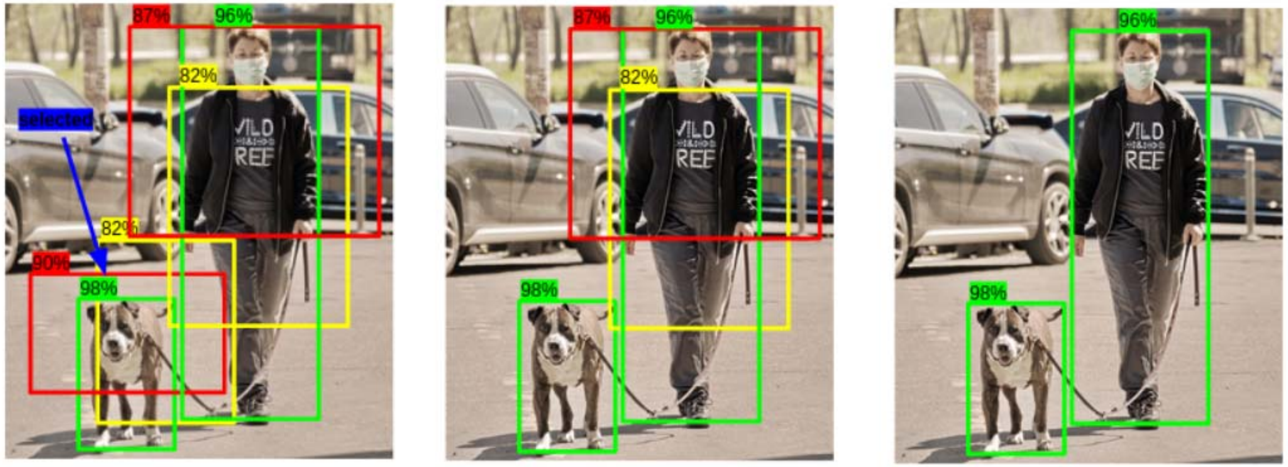
IoU threshold = 0.5 = *False Positive (FP)*

IoU threshold = 0.2 = *True Positive (TP)*

https://towardsdatascience.com/map-mean-average-precision-might-confuse-you-5956f1bfa9e2

YOLO (You Only Look Once)

- NMS (**Non Maximum Suppression**)
  - The process of selecting the best bounding box using NMS
  - Step 1: Select the box with the highest objectiveness score
  - Step 2: Then compare the overlap (IOU) of this box with other boxes
  - Step 3: Remove bounding boxes with IOUs greater than 50%
  - Step 4: Go to the next higher objectiveness score
  - Step 5: Repeat steps 2-4.



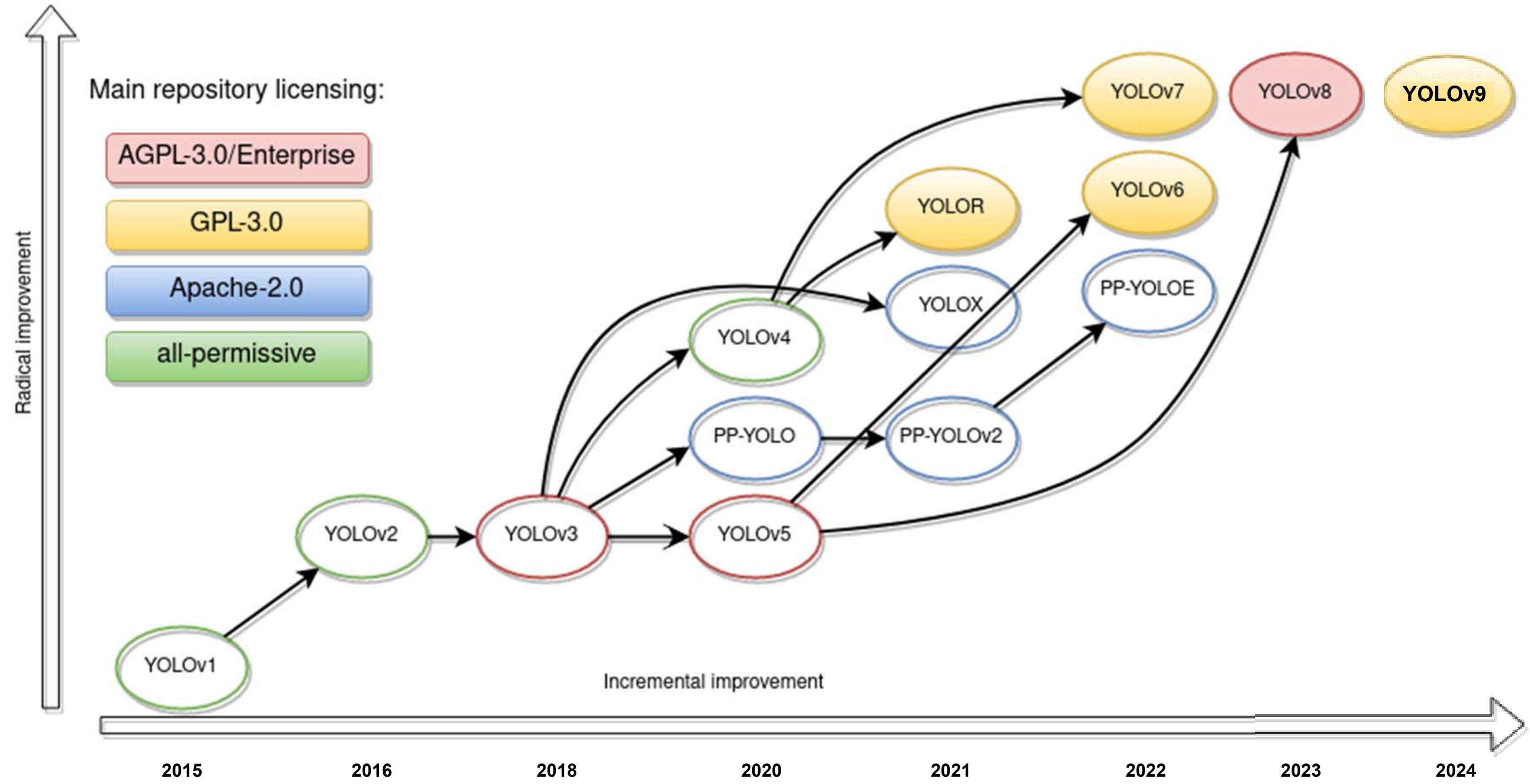
Step 1: Selecting Bounding box with highest score

Step 3: Delete Bounding box with high overlap

Step 5: Final Output

YOLO (You Only Look Once)

- YOLO Family

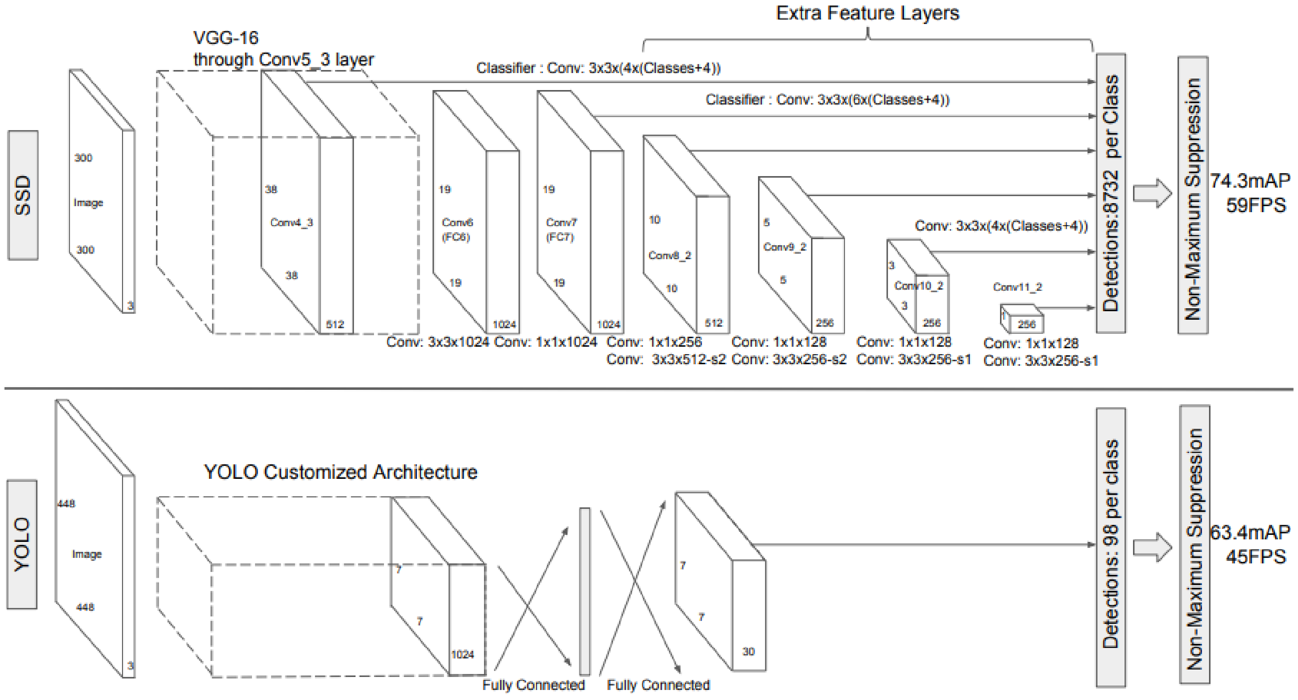

Main repository licensing:
AGPL-3.0/Enterprise
GPL-3.0
Apache-2.0
all-permissive
Radical improvement
Incremental improvement
YOLOv1
YOLOv2
YOLOv3
YOLOv4
PP-YOLO
YOLOv5
YOLOR
YOLOX
PP-YOLOv2
YOLOv7
YOLOv6
PP-YOLOE
YOLOv8
YOLOv9
2015
2016
2018
2020
2021
2022
2023
2024

https://learnopencv.com/mastering-all-yolo-models/

https://www.linkedin.com/pulse/evolution-yolo-neural-network-family-from-v1-v8-part-2-max-ivanov

SSD (Single Shot MultiBox Detector)

- It is one of the first attempts at using convolutional neural network's pyramidal feature hierarchy for efficient detection of objects of various sizes.

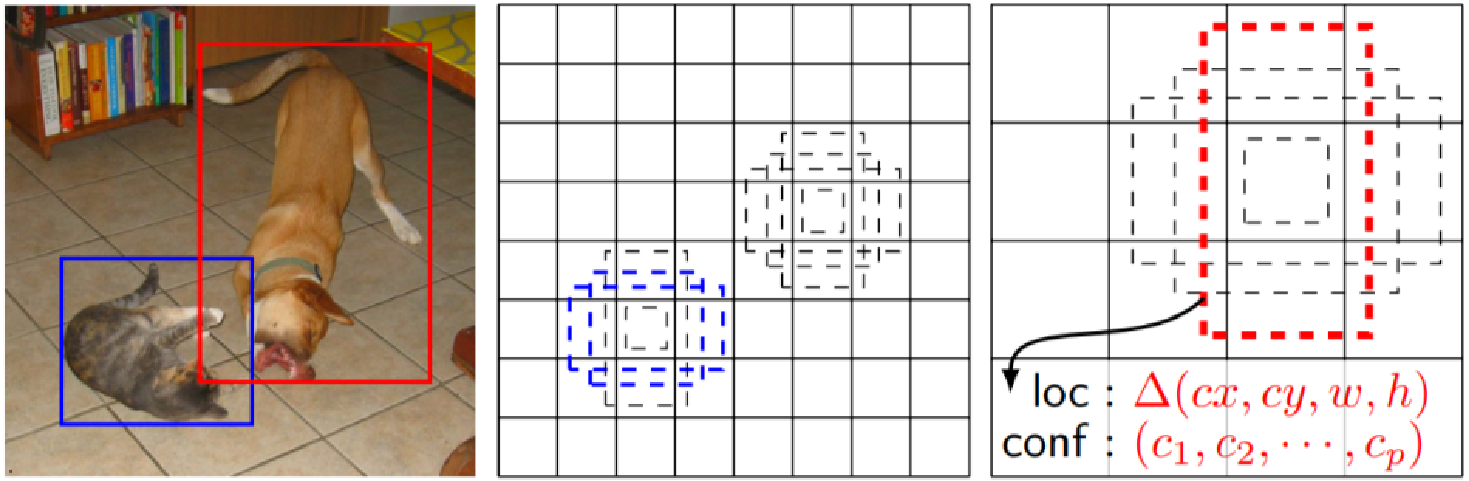

Extra Feature Layers
VGG-16
through Conv5_3 layer
Classifier : Conv: 3x3x(4x(Classes+4))
Classifier : Conv: 3x3x(6x(Classes+4))
SSD
300
Image
300
3
38
Conv4_3
38
512
19
Conv6
(FC6)
19
1024
19
Conv7
(FC7)
19
1024
10
Conv8_2
10
512
5
Conv9_2
5
256
3
Conv10_2
3
256
Conv11_2
1
256
Conv: 3x3x(4x(Classes+4))
Conv: 3x3x1024
Conv: 1x1x1024
Conv: 1x1x256
Conv: 3x3x512-s2
Conv: 1x1x128
Conv: 3x3x256-s2
Conv: 1x1x128
Conv: 3x3x256-s1
Conv: 1x1x128
Conv: 3x3x256-s1
Detections:8732 per Class
Non-Maximum Suppression
74.3mAP
59FPS
YOLO
448
Image
448
3
YOLO Customized Architecture
7
7
1024
Fully Connected
Fully Connected
7
7
30
Detections: 98 per class
Non-Maximum Suppression
63.4mAP
45FPS

(Paper) SSD: Single Shot MultiBox Detector

SSD (Single Shot MultiBox Detector)

- Multi-scale and Anchor box (Default box)

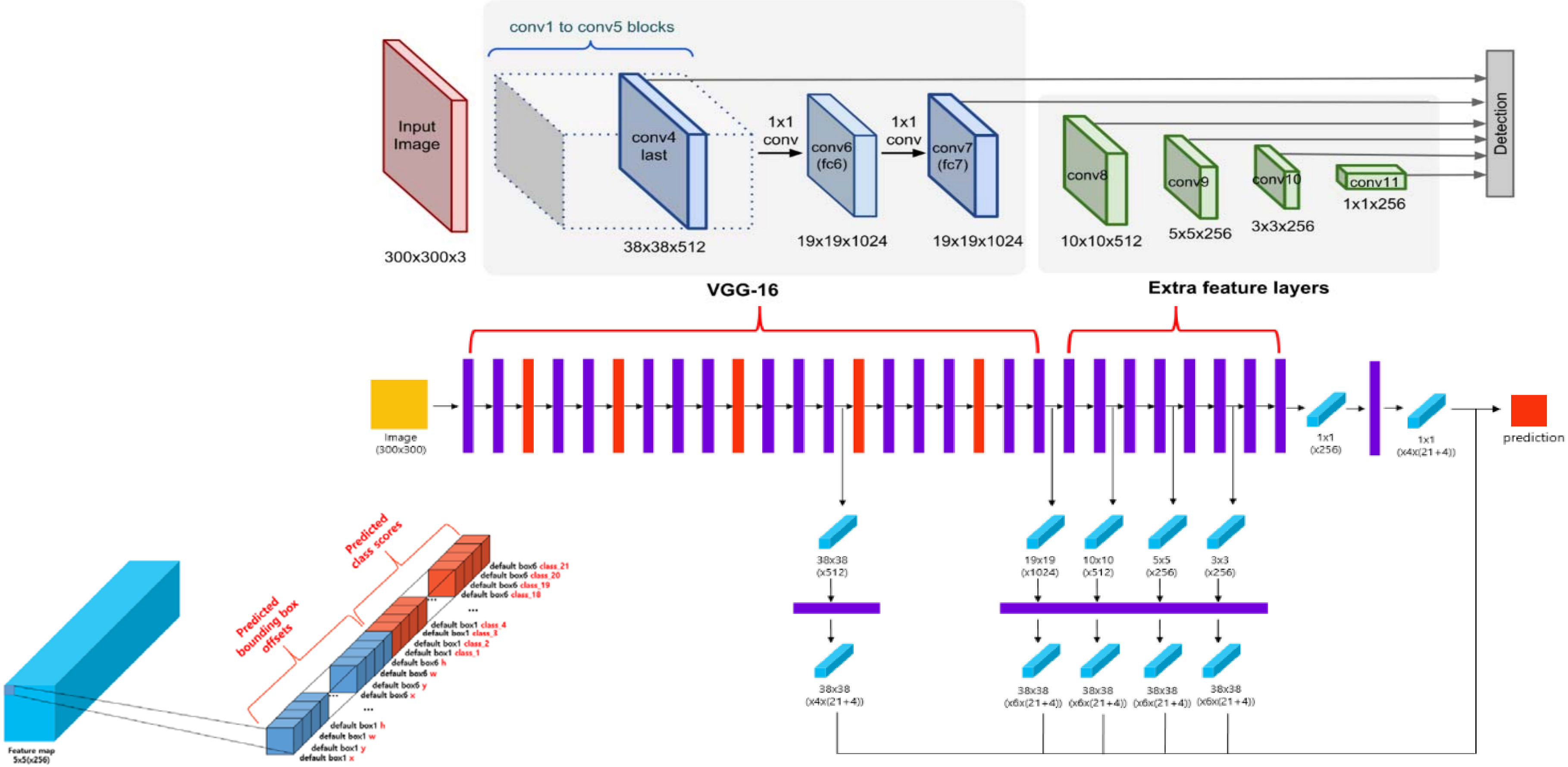
$$s_k = s_{min} + \frac{s_{max} - s_{min}}{m-1}(k-1), k \in [1, m]$$


loc : $\Delta(cx, cy, w, h)$
conf : $(c_1, c_2, \cdots, c_p)$

(a) Image with GT boxes (b) 8 × 8 feature map (c) 4 × 4 feature map

SSD (Single Shot MultiBox Detector)

- Default boxes & Prediction

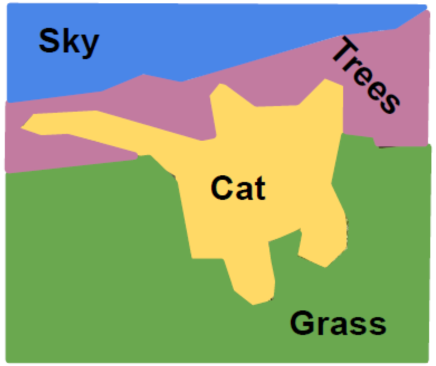

conv1 to conv5 blocks
Input Image
300x300x3
conv4 last
38x38x512
1x1 conv
conv6 (fc6)
19x19x1024
1x1 conv
conv7 (fc7)
19x19x1024
conv8
10x10x512
conv9
5x5x256
conv10
3x3x256
conv11
1x1x256
Detection
VGG-16
Extra feature layers
Image (300x300)
1x1 (x256)
1x1 (x4x(21+4))
prediction
38x38 (x512)
19x19 (x1024)
10x10 (x512)
5x5 (x256)
3x3 (x256)
38x38 (x4x(21+4))
38x38 (x6x(21+4))
38x38 (x6x(21+4))
38x38 (x6x(21+4))
38x38 (x6x(21+4))
Feature map 5x5(x256)
Predicted bounding box offsets
Predicted class scores
default box6 class_21
default box6 class_20
default box6 class_19
default box6 class_18
...
default box1 class_4
default box1 class_3
default box1 class_2
default box1 class_1
default box6 h
default box6 w
default box6 y
default box6 x
...
default box1 h
default box1 w
default box1 y
default box1 x

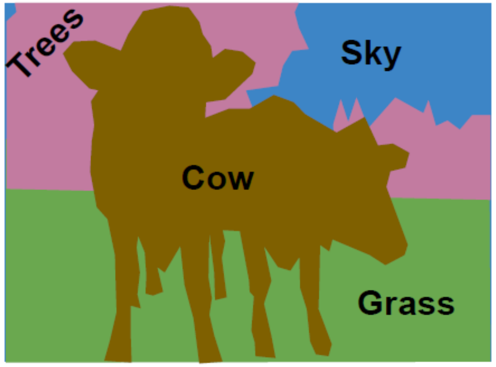
Segmentation

(FCN / Mask-RCNN)

Semantic Segmentation

- Label each pixel in the image with a category label
- Don’t differentiate instances, only care about pixels


Sky
Trees
Cat
Grass


Trees
Sky
Cow
Grass

Semantic Segmentation

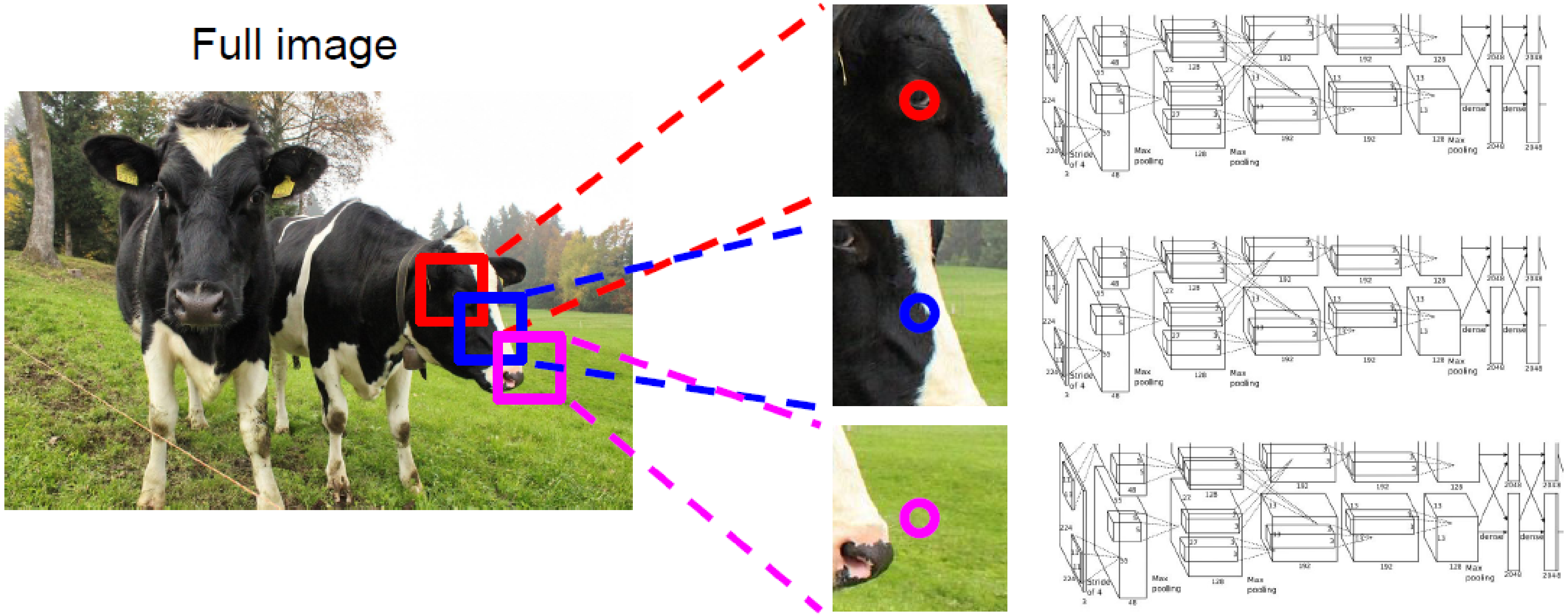
- Sliding Window

Full image

Extract patch

Classify center pixel with CNN

What's problem?



Cow

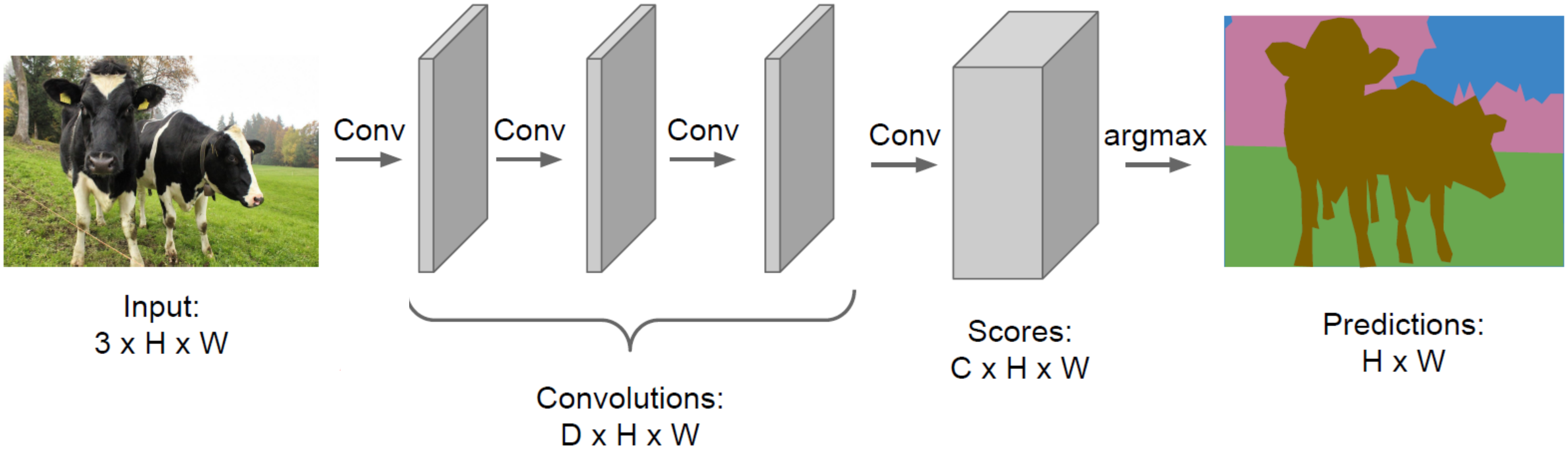
Cow

Grass

Farabet et al, "Learning Hierarchical Features for Scene Labeling," TPAMI 2013
Pinheiro and Collobert, "Recurrent Convolutional Neural Networks for Scene Labeling", ICML 2014

Semantic Segmentation

- Fully Convolutional Network
  - Design a network as a bunch of convolutional layers to make predictions for pixels all at once

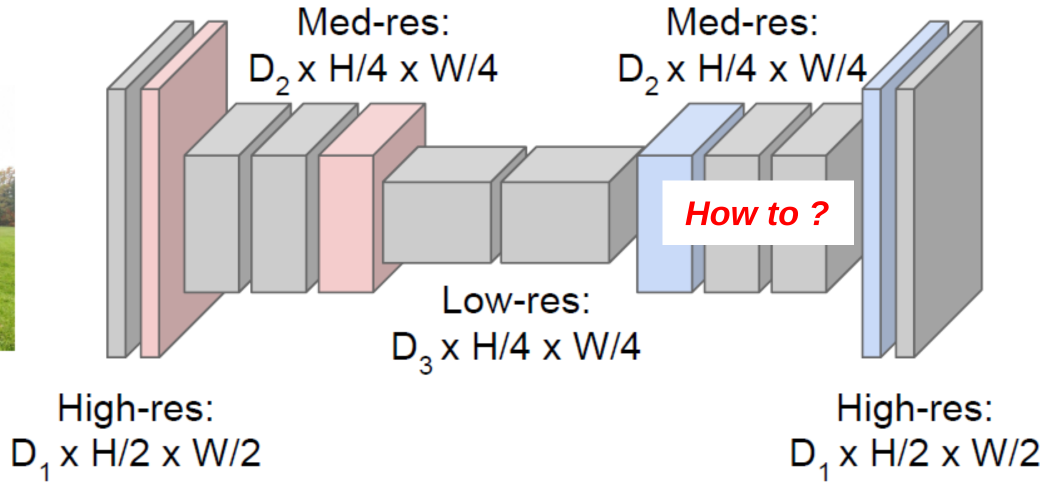

Conv
Conv
Conv
Conv
argmax
Input:
3 x H x W
Convolutions:
D x H x W
Scores:
C x H x W
Predictions:
H x W

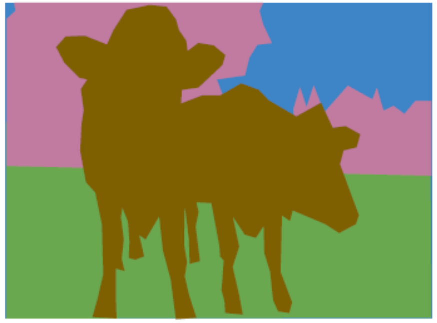
What's problem?

Semantic Segmentation

- Fully Convolutional Network (Encoder & Decoder)
  - Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!

Input:
3 x H x W


Med-res:
D_2 x H/4 x W/4
Med-res:
D_2 x H/4 x W/4
How to ?
Low-res:
D_3 x H/4 x W/4
High-res:
D_1 x H/2 x W/2
High-res:
D_1 x H/2 x W/2

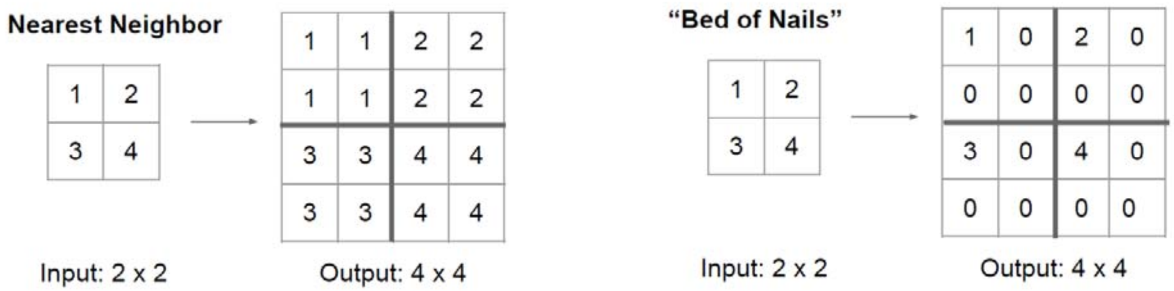


Predictions:
H x W

Long, Shelhamer, and Darrell, “Fully Convolutional Networks for Semantic Segmentation”, CVPR 2015
Noh et al, “Learning Deconvolution Network for Semantic Segmentation”, ICCV 2015

Semantic Segmentation

- In-Network upsampling: **Unpooling**



Nearest Neighbor
→
Input: 2 x 2
Output: 4 x 4
"Bed of Nails"
→
Input: 2 x 2
Output: 4 x 4

Nearest Neighbor, Input: 2 x 2

1	2
3	4

Output: 4 x 4

1	1	2	2
1	1	2	2
3	3	4	4
3	3	4	4

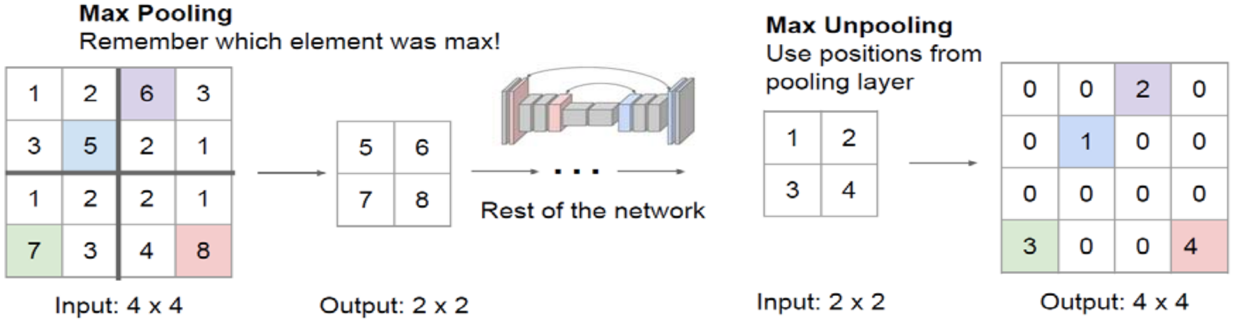
"Bed of Nails", Input: 2 x 2

1	2
3	4

Output: 4 x 4

1	0	2	0
0	0	0	0
3	0	4	0
0	0	0	0

- In-Network upsampling: **Max Unpooling**
  - Corresponding pairs of downsampling and upsampling layers



Max Pooling
Remember which element was max!
→
Input: 4 x 4
Output: 2 x 2
→ . . . →
Rest of the network
Max Unpooling
Use positions from pooling layer
→
Input: 2 x 2
Output: 4 x 4

Max Pooling, Input: 4 x 4

1	2	6	3
3	5	2	1
1	2	2	1
7	3	4	8

Output: 2 x 2

5	6
7	8

Max Unpooling, Input: 2 x 2

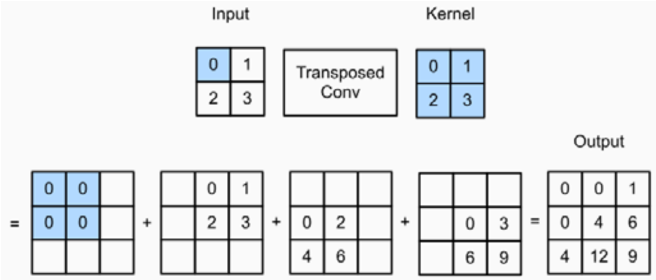
1	2
3	4

Output: 4 x 4

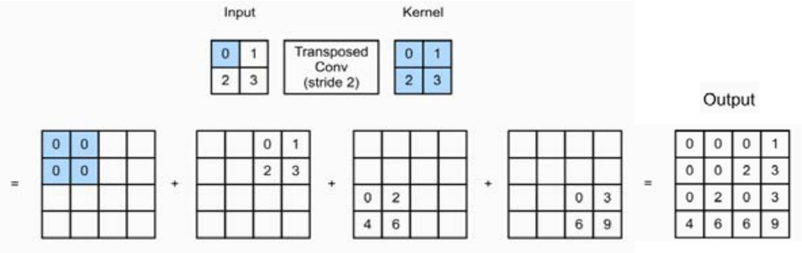
0	0	2	0
0	1	0	0
0	0	0	0
3	0	0	4

Semantic Segmentation

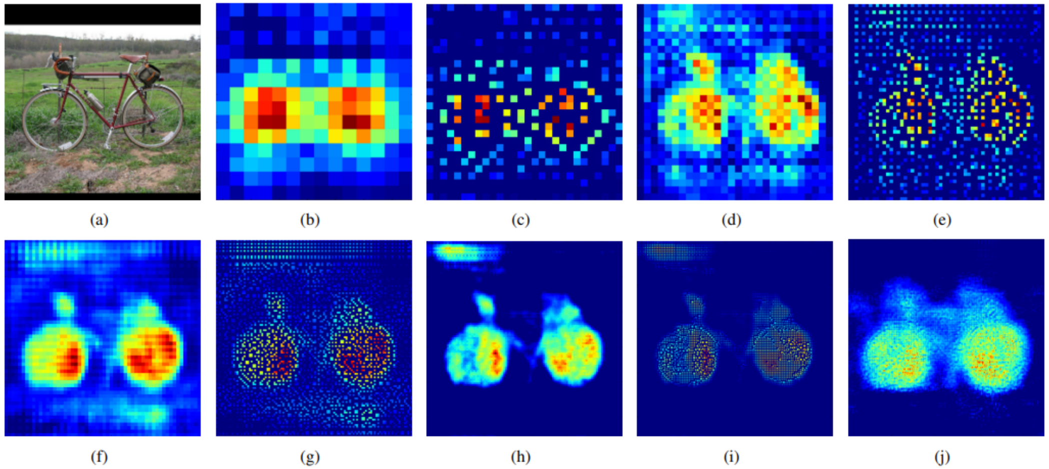
- Learnable Upsampling: **Transpose Convolution**
  - In contrast to the regular convolution that reduces input elements via the kernel, the transposed convolution broadcasts input elements via the kernel, thereby producing an ***output that is larger than the input***


Input
Kernel
Transposed Conv
Output

Transposed convolution with a 2×2 kernel (Basic Operation)


Input
Kernel
Transposed Conv (stride 2)
Output

Transposed convolution with a 2×2 kernel with stride of 2

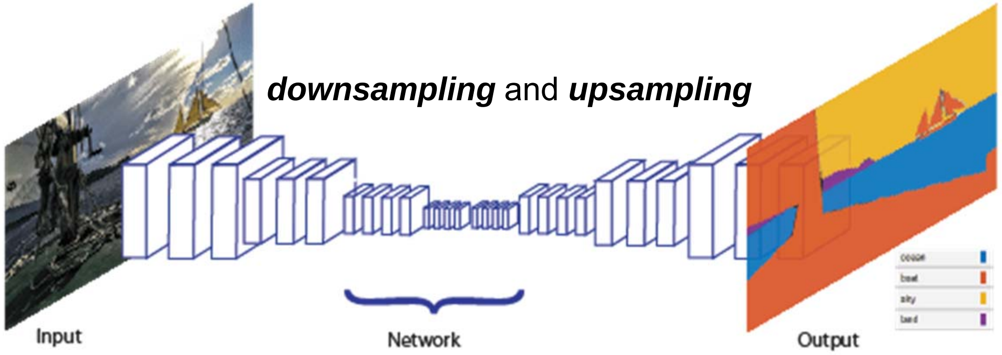

(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
(j)

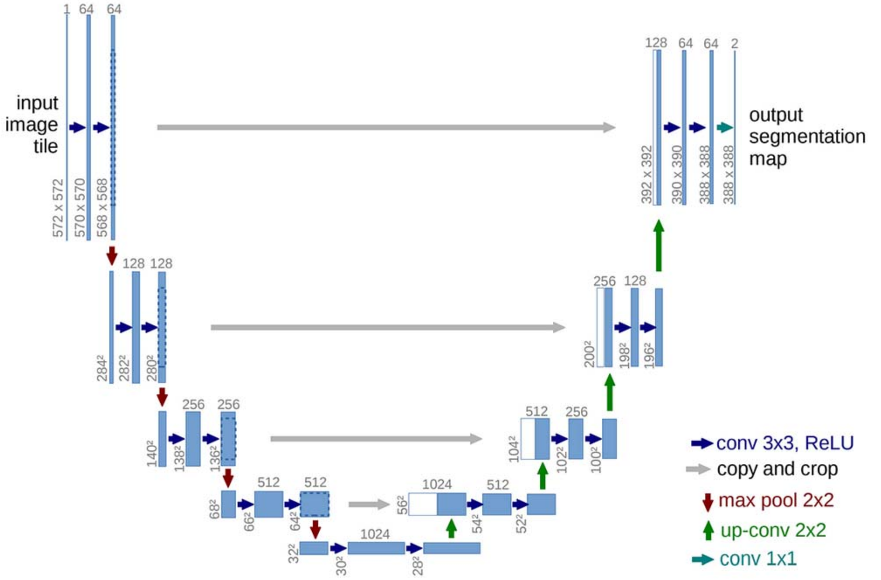
Activation maps

Learning Deconvolution Network for Semantic Segmentation, CVPR 2015

Semantic Segmentation

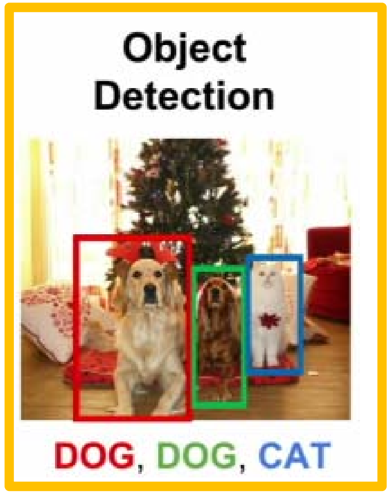
- Fully Convolutional Network
  - Downsampling: Pooling, strided convolution
  - Upsampling: Unpooling or strided transpose convolution

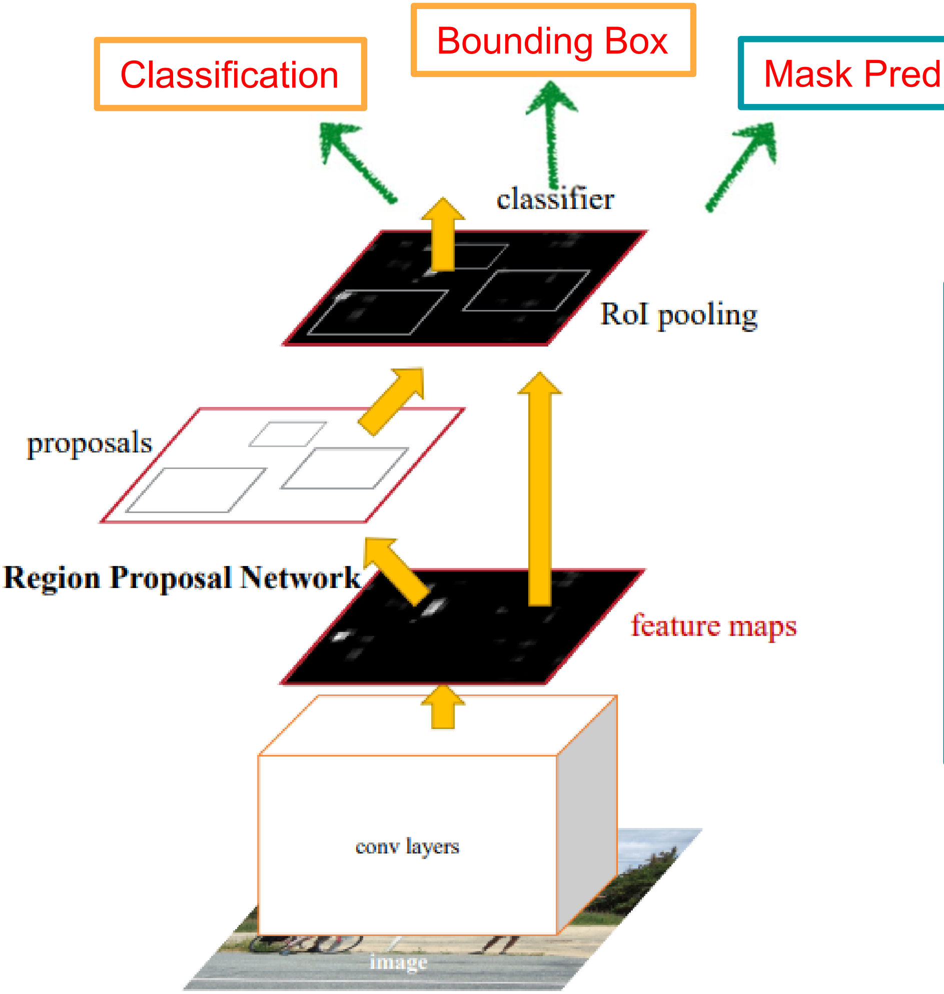

downsampling and *upsampling*
Input
Network
Output
ocean
boat
sky
land


1 64 64
input image tile
572 x 572
570 x 570
568 x 568
128 128
284^2
282^2
280^2
256 256
140^2
138^2
136^2
512 512
68^2
66^2
64^2
1024
32^2
30^2
28^2
1024
56^2
512
54^2
52^2
512 256
104^2
102^2
100^2
256 128
200^2
198^2
196^2
128 64 64 2
392 x 392
390 x 390
388 x 388
388 x 388
output segmentation map
conv 3x3, ReLU
copy and crop
max pool 2x2
up-conv 2x2
conv 1x1

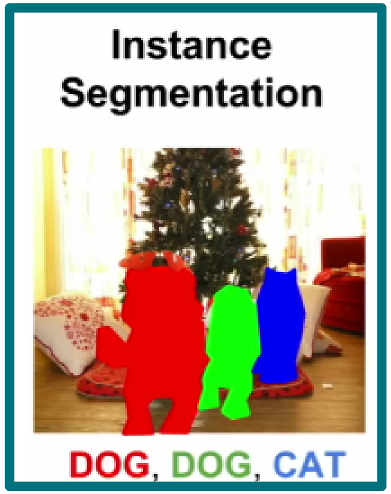
U-Net : Convolutional Networks for Biomedical Image Segmentation, MICCAI 2015

Instance Segmentation


Object Detection
DOG, DOG, CAT


Classification
Bounding Box
Mask Prediction
classifier
RoI pooling
proposals
Region Proposal Network
feature maps
conv layers
image

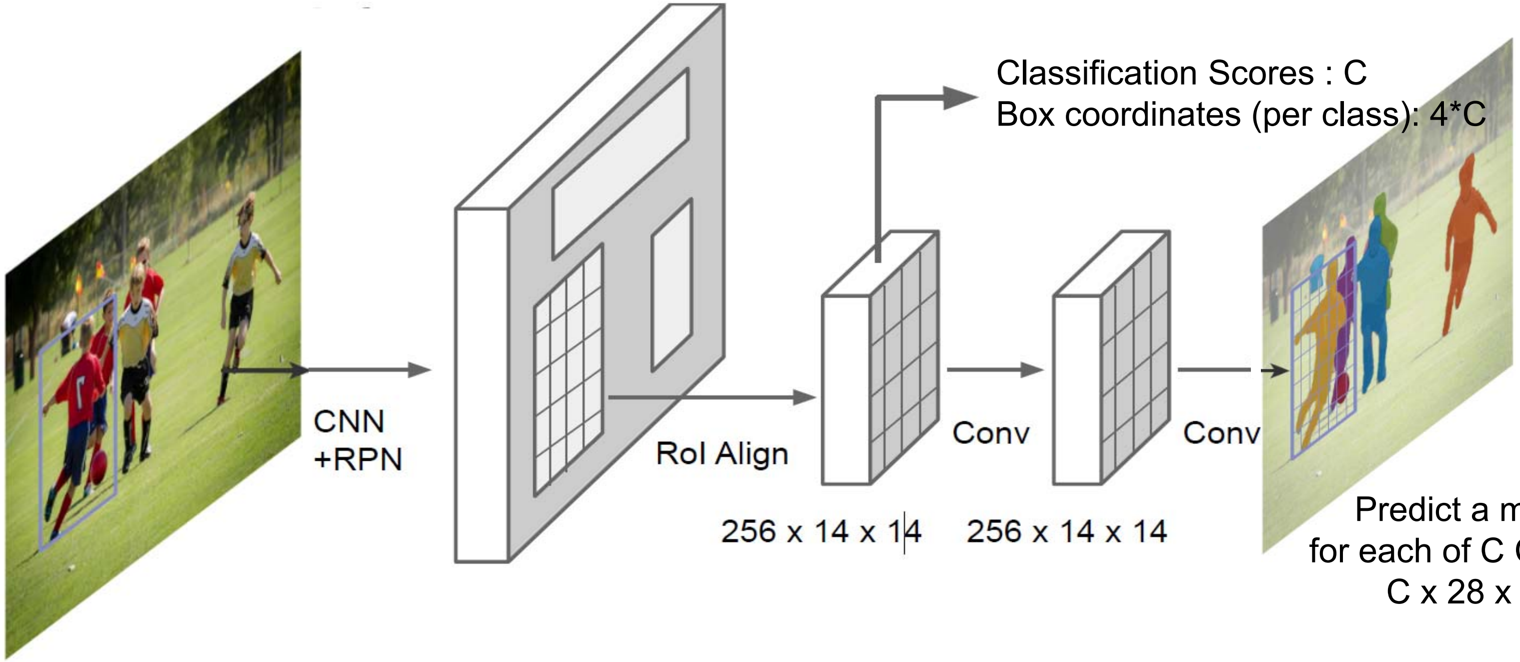
Add a small mask network that operates on each RoI and predicts a 28x28 binary mask


Instance Segmentation
DOG, DOG, CAT

He et al, “Mask R-CNN”, ICCV 2017

Mask R-CNN

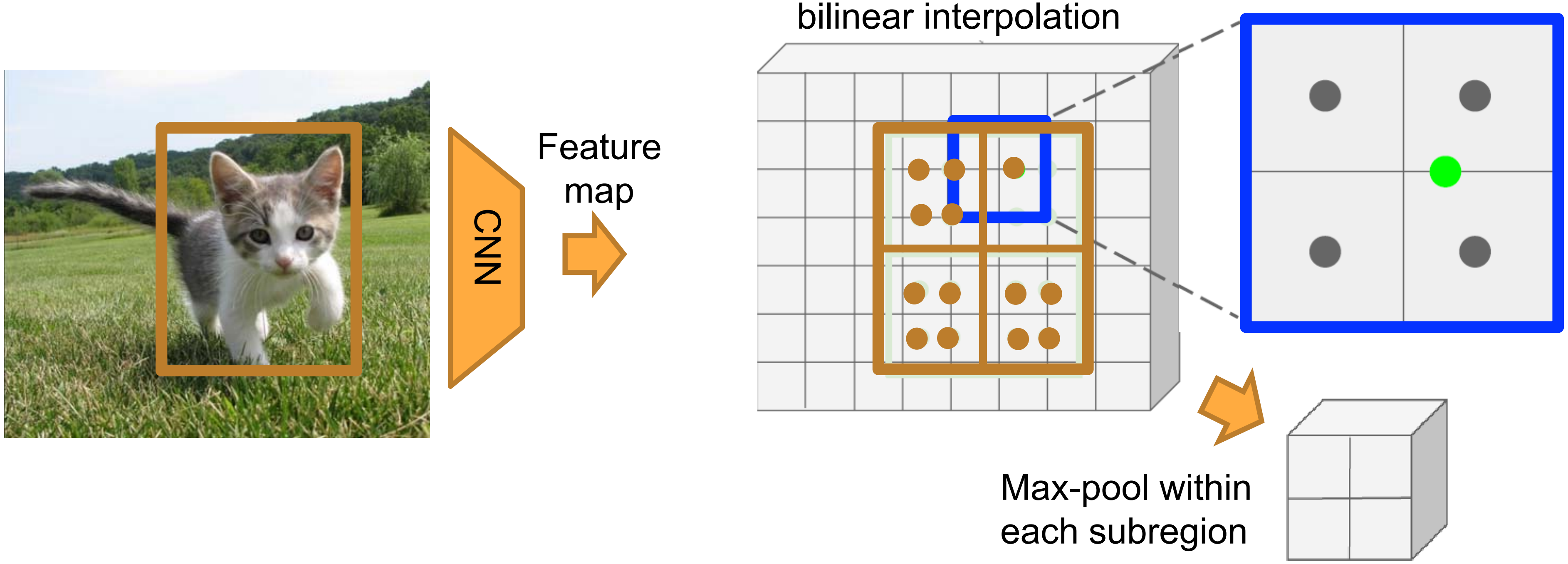
- Faster R-CNN + FCN (Mask branch)
- RoI Pooling → RoI Align
- Decouple class and mask prediction
- Multi-task Loss Function


Classification Scores : C
Box coordinates (per class): 4*C
CNN
+RPN
RoI Align
Conv
Conv
256 x 14 x 14
256 x 14 x 14
Predict a mask
for each of C Classes
C x 28 x 28

Mask R-CNN

- **RoI Align**

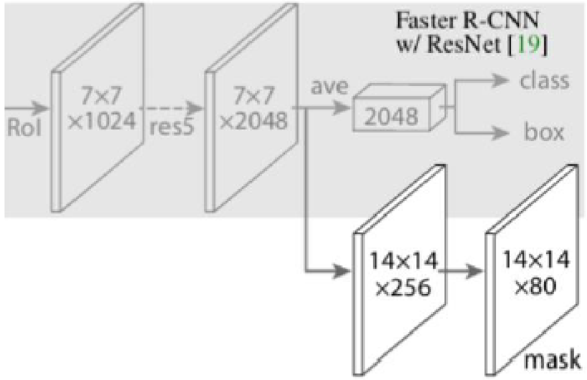
(1) No “Snap”
(2) Sample at regular points in each subregion using bilinear interpolation

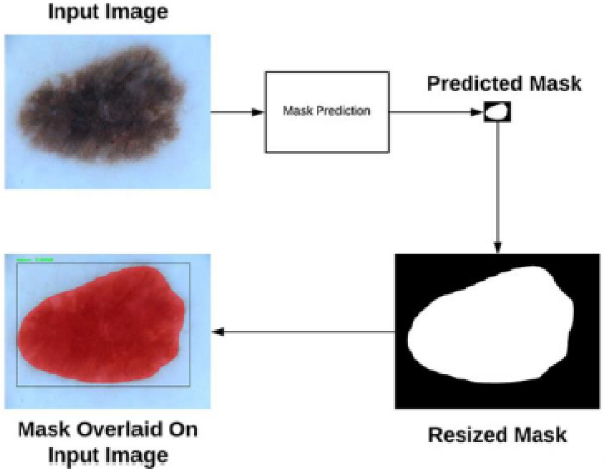


Max-pool within each subregion

Mask R-CNN

- Mask Prediction


Faster R-CNN
w/ ResNet [19]
RoI
7×7
×1024
res5
7×7
×2048
ave
2048
class
box
14×14
×256
14×14
×80
mask


Input Image

Mask Prediction

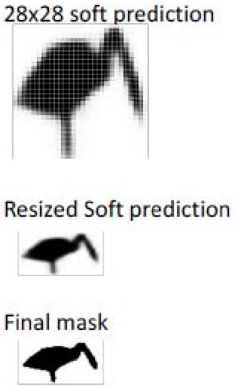
Predicted Mask

Resized Mask

Mask Overlaid On Input Image

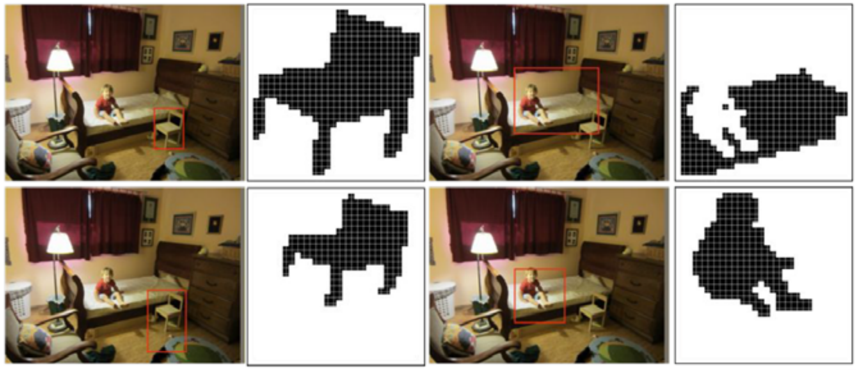
Multi-task Loss Function

$$L = L_{cls} + L_{box} + \boxed{L_{mask}}$$


28x28 soft prediction

Resized Soft prediction

Final mask


Mask Training Target

Data Set / Evaluation

Data Set

http://host.robots.ox.ac.uk/pascal/VOC/

- **Pascal VOC**(Visual Object Classes)
  - Provides standardised image data sets for object class recognition
  - Provides a common set of tools for accessing the data sets and annotations
  - Enables evaluation and comparison of different methods
  - Ran challenges evaluating performance on object class recognition (from 2005-2012, now finished)
- Format
  - XML Format
  - 20 Categories

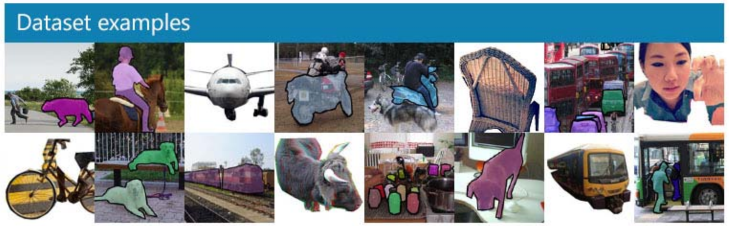
- **VOCdevkit**
  - **VOC2012**
    - **Annotations** : xml format / Annotation for each image
    - **ImageSet** : train, test, trainval, val
    - **JPEGImages** : Image for Detectin & Segmentation
    - **SegmentationClass** : Masking Image for Semantic Segmentation
    - **SegmentationObject** : Masking Image for Instance Segmentation

Data Set

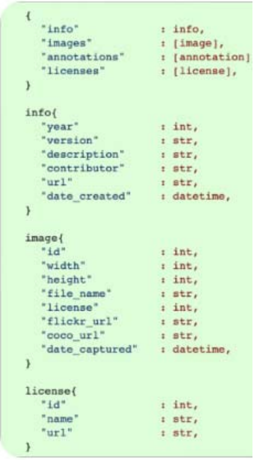
Site : https://cocodataset.org/#home
Paper : https://arxiv.org/pdf/1405.0312

- **MS COCO** (Common Objects in Context)
  - COCO is a large-scale object detection, segmentation, and captioning dataset
  - COCO has several features:

✓ Object segmentation
✓ Recognition in context
✓ Superpixel stuff segmentation
✓ 330K images (>200K labeled)
✓ 1.5 million object instances
✓ 80 object categories
✓ 91 stuff categories
✓ 5 captions per image
✓ 250,000 people with keypoints


Dataset examples

  - json Formet



```
{
   "info"              : info,
   "images"            : [image],
   "annotations"       : [annotation]
   "licenses"          : [license],
}

info{
   "year"              : int,
   "version"           : str,
   "description"       : str,
   "contributor"       : str,
   "url"               : str,
   "date_created"      : datetime,
}

image{
   "id"                : int,
   "width"             : int,
   "height"            : int,
   "file_name"         : str,
   "license"           : int,
   "flickr_url"        : str,
   "coco_url"          : str,
   "date_captured"     : datetime,
}

license{
   "id"                : int,
   "name"              : str,
   "url"               : str,
}
```

info : general information about the dataset, such as version number, date created, and contributor information

license : information about the licenses for the images in the dataset

images : a list of all the images in the dataset, including the file path, width, height, and other metadata

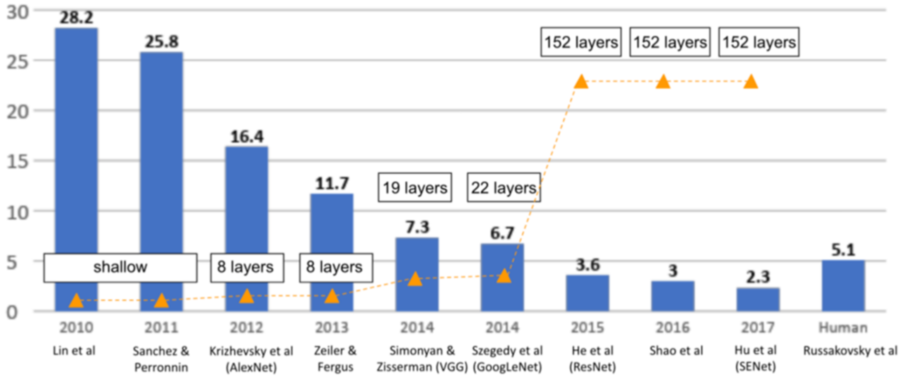
annotations : a list of all the object annotations for each image, including the object category, bounding box coordinates, and segmentation masks (if available)

categories : a list of all the dataset object categories,

Data Set

https://www.image-net.org/

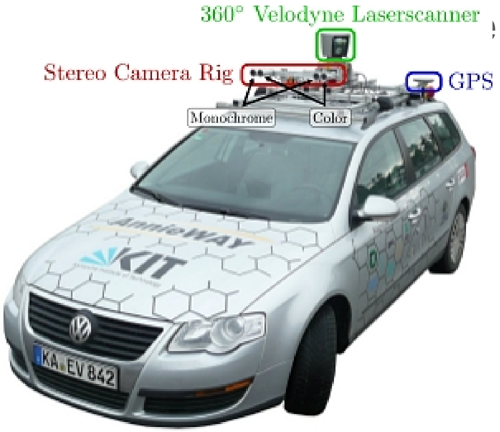
- **ImageNet**
  - The ImageNet Large Scale Visual Recognition Challenge (ILSVRC) evaluates algorithms for object detection and image classification at large scale
  - It contains over 14 million high-resolution images spanning thousands of object categories.
  - The dataset is organized according to the WordNet hierarchy
  - ImageNet is widely used for training and benchmarking in the field of computer vision
  - The annual ILSVRC has been instrumental in advancing computer vision research (2010~2017)



Data Set

https://www.cvlibs.net/datasets/kitti/

- **KITTI** (Karlsruhe Institute of Technology and Toyota Technological Institute at Chicago)
  - To develop novel challenging real-world computer vision benchmarks
  - Tasks of interest are: stereo, optical flow, visual odometry, 3D object detection and 3D tracking
  - Datasets are captured by driving around the mid-size city of Karlsruhe, in rural areas and on highways. Up to 15 cars and 30 pedestrians are visible per image
  - All datasets and benchmarks of KITTI are copyright by us and publi Commons Attribution-NonCommercial-ShareAlike 3.0 Liecens

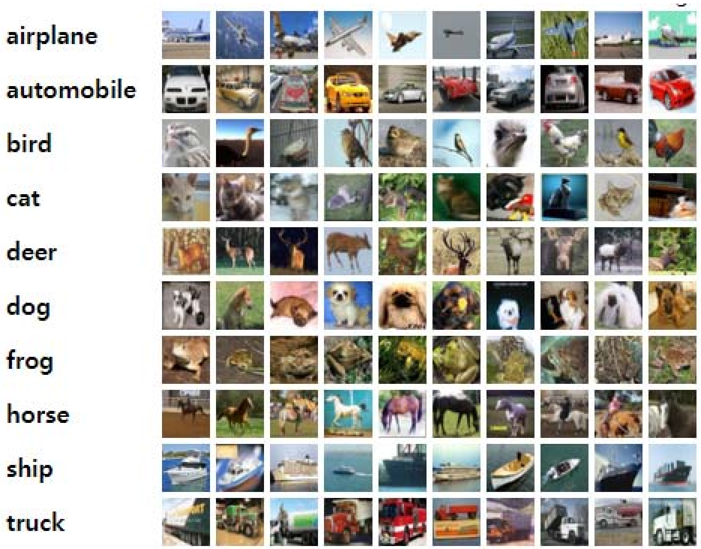


Data Set

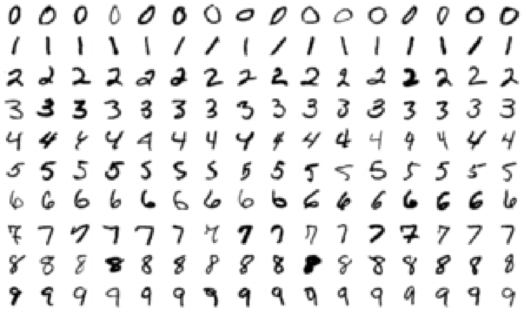
https://www.cs.toronto.edu/~kriz/cifar.html

- **CIFAR-10/100** (Canadian Institute for Advanced Research, 10/100 classes)
  - CIFAR-10 dataset consists of 60000 32x32 colour images in 10 classes, with 6000 images per class. There are 50000 training images and 10000 test images
  - CIFAR-100 has 100 classes containing 600 images each. There are 500 training images and 100 testing images per class
    - The 100 classes in the CIFAR-100 are grouped into 20 superclasses

Superclass	**Classes**
aquatic mammals	beaver, dolphin, otter, seal, whale
fish	aquarium fish, flatfish, ray, shark, trout
flowers	orchids, poppies, roses, sunflowers, tulips
food containers	bottles, bowls, cans, cups, plates
fruit and vegetables	apples, mushrooms, oranges, pears, sweet peppers
household electrical devices	clock, computer keyboard, lamp, telephone, television
household furniture	bed, chair, couch, table, wardrobe
insects	bee, beetle, butterfly, caterpillar, cockroach
⋮	⋮

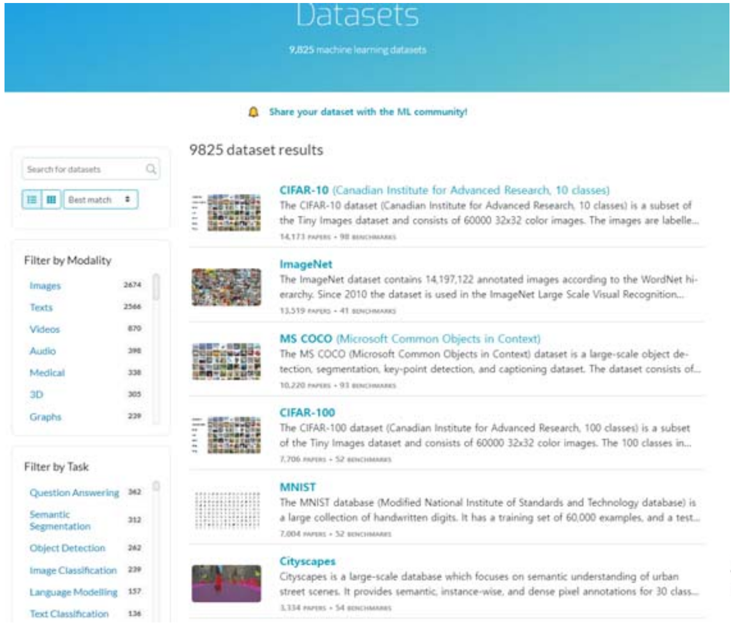


Data Set



- **MNIST** (Modified National Institute of Standards and Technology)
  - MNIST database of handwritten digits
  - It has a training set of 60,000 examples, and a test set of 10,000 examples
    http://yann.lecun.com/exdb/mnist/

- **Others**
  - The are about 10,000 machine learning datasets
  - https://paperswithcode.com/datasets


Datasets
9,825 machine learning datasets
Share your dataset with the ML community!
Search for datasets
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Filter by Modality
Images 2674
Texts 2566
Videos 870
Audio 398
Medical 338
3D 305
Graphs 239
Filter by Task
Question Answering 362
Semantic Segmentation 312
Object Detection 262
Image Classification 239
Language Modelling 157
Text Classification 136

9825 dataset results

CIFAR-10 (Canadian Institute for Advanced Research, 10 classes)
The CIFAR-10 dataset (Canadian Institute for Advanced Research, 10 classes) is a subset of the Tiny Images dataset and consists of 60000 32x32 color images. The images are labelle...
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ImageNet
The ImageNet dataset contains 14,197,122 annotated images according to the WordNet hierarchy. Since 2010 the dataset is used in the ImageNet Large Scale Visual Recognition...
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MS COCO (Microsoft Common Objects in Context)
The MS COCO (Microsoft Common Objects in Context) dataset is a large-scale object detection, segmentation, key-point detection, and captioning dataset. The dataset consists of...
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CIFAR-100
The CIFAR-100 dataset (Canadian Institute for Advanced Research, 100 classes) is a subset of the Tiny Images dataset and consists of 60000 32x32 color images. The 100 classes in...
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MNIST
The MNIST database (Modified National Institute of Standards and Technology database) is a large collection of handwritten digits. It has a training set of 60,000 examples, and a test...
7,004 PAPERS • 52 BENCHMARKS

Cityscapes
Cityscapes is a large-scale database which focuses on semantic understanding of urban street scenes. It provides semantic, instance-wise, and dense pixel annotations for 30 class...
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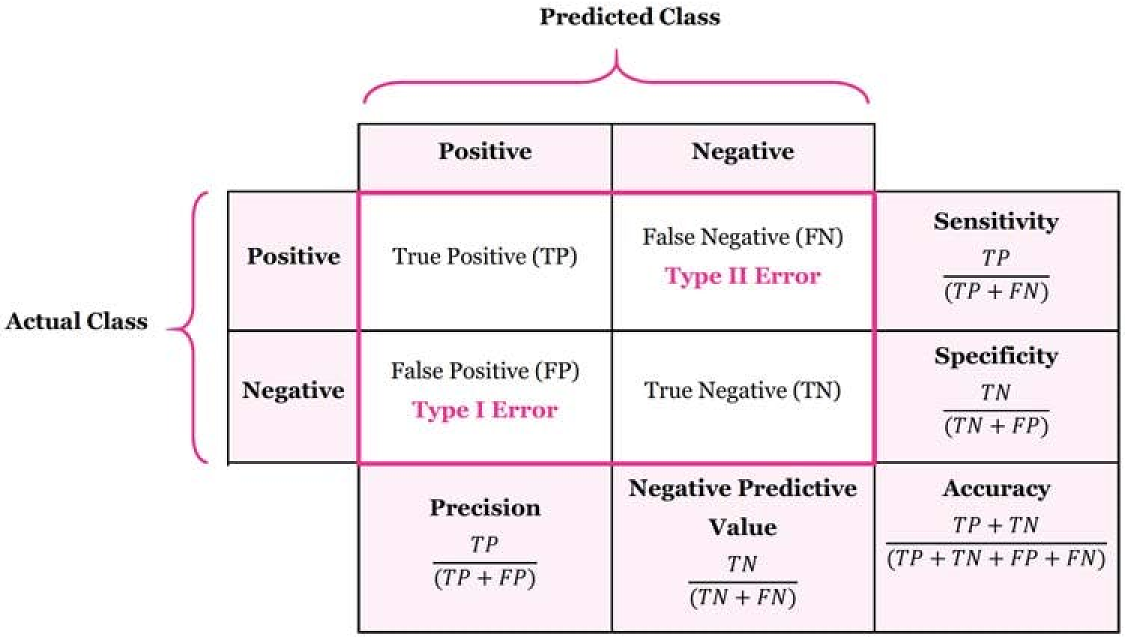
Evaluation

Confusion Matrix

- A confusion matrix is a performance evaluation tool used in machine learning that summarizes the performance of a classification model by tabulating true positive, true negative, false positive, and false negative predictions

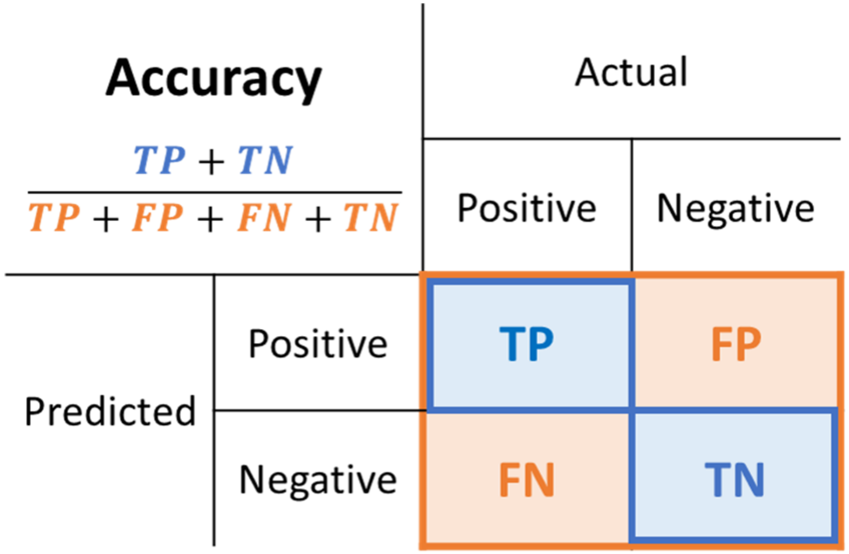


		Predicted Class		
		Positive	**Negative**	
Actual Class	**Positive**	True Positive (TP)	False Negative (FN) **Type II Error**	**Sensitivity** $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP) **Type I Error**	True Negative (TN)	**Specificity** $\frac{TN}{(TN+FP)}$
		Precision $\frac{TP}{(TP+FP)}$	**Negative Predictive Value** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

- **True Positives (TP)**: Instances where the model correctly predicts a positive class when it is indeed positive. Consider a cancer diagnostic model: a true positive would occur when the model correctly identifies a patient with cancer as having the disease.
- **True Negatives (TN)**: Instances where the model correctly predicts a negative class when it is indeed negative. Continuing the medical analogy, a true negative would be when the model correctly identifies a healthy patient as not having the disease.
- **False Positives (FP)**: Instances where the model incorrectly predicts a positive class when it should have been negative. In the medical scenario, a false positive would mean the model wrongly indicates a patient has the disease when they are, in fact, healthy.
- **False Negatives (FN):** Instances where the model incorrectly predicts a negative class when it should have been positive. In the medical context, a false negative would be when the model fails to detect a disease in a patient who actually has it.

Evaluation

- Accuracy
  - Accuracy quantifies the ratio of correct predictions (TP and TN) to the total number of predictions. While informative, this metric can be misleading when classes are imbalanced.



Accuracy $\frac{TP + TN}{TP + FP + FN + TN}$		Actual	
		Positive	Negative
Predicted	Positive	TP	FP
	Negative	FN	TN

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

Hospital A

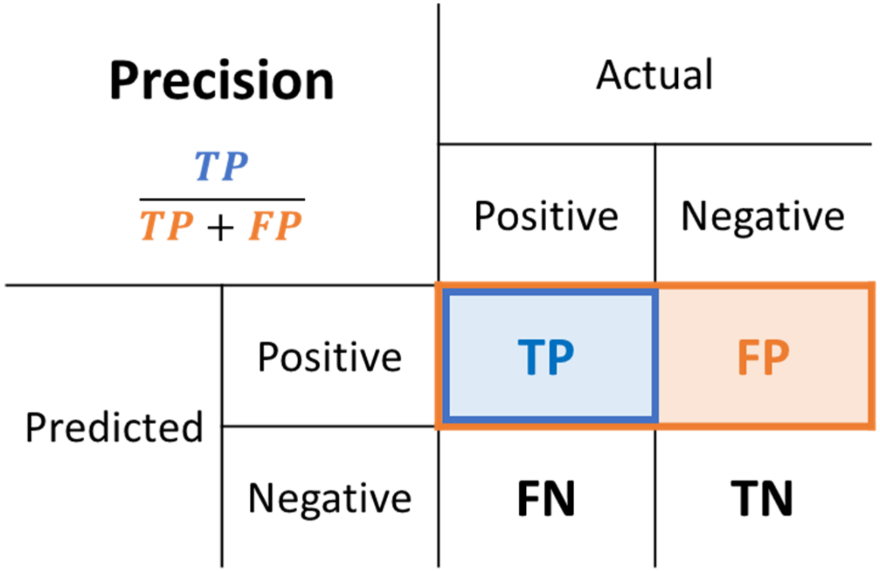
Actual	**Healthy**	**Healthy**	**Healthy**	**Healthy**	**Cancer**
Predicted	Healthy	Healthy	Healthy	Cancer	Healthy
	O	O	O	X	X

Hospital B

Actual	**Healthy**	**Healthy**	**Healthy**	**Healthy**	**Cancer**
Predicted	Healthy	Healthy	Cancer	Cancer	Cancer
	O	O	X	X	O

Evaluation

- Precision
  - Precision evaluates the proportion of true positive predictions among all positive predictions (TP / (TP + FP)). This metric is crucial when the cost of false positives is high



Precision

$$\frac{TP}{TP + FP}$$

		Actual	
		Positive	Negative
Predicted	Positive	TP	FP
	Negative	FN	TN

$$\text{Precision} = \frac{TP}{TP + FP}$$

Hospital A

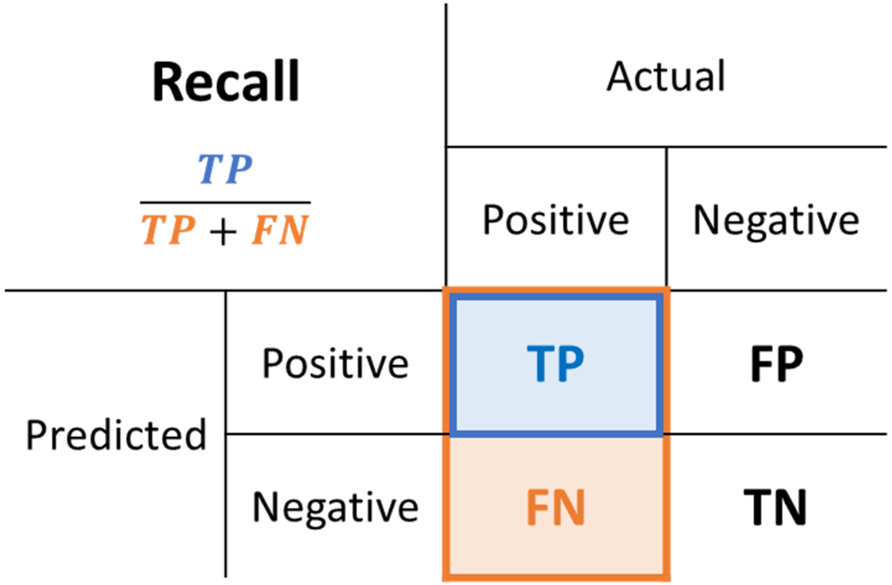
Actual	**Healthy**	**Healthy**	**Healthy**	**Healthy**	**Cancer**
Predicted	Healthy	Healthy	Healthy	Cancer	Healthy
	O	O	O	X	X

Hospital B

Actual	**Healthy**	**Healthy**	**Healthy**	**Healthy**	**Cancer**
Predicted	Healthy	Healthy	Cancer	Cancer	Cancer
	O	O	X	X	O

Evaluation

- Recall
  - Recall measures the ratio of true positive predictions to the actual number of positive instances (TP / (TP + FN)). This metric is significant when missing positive instances is costly.
  - It called Sensitivity or True Positive Rate



Recall

$$\frac{TP}{TP + FN}$$

Recall		Actual	
		Positive	Negative
Predicted	Positive	**TP**	**FP**
	Negative	**FN**	**TN**

$$\text{Recall} = \frac{TP}{TP + FN}$$

Hospital A

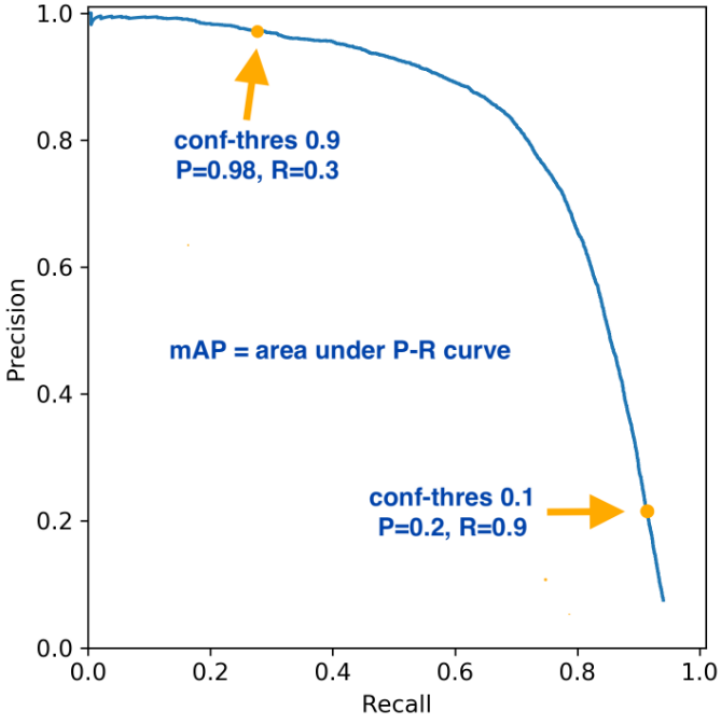
Actual	**Healthy**	**Healthy**	**Healthy**	**Healthy**	**Cancer**
Predicted	Healthy	Healthy	Healthy	Cancer	Healthy
	O	O	O	X	X

Hospital B

Actual	**Healthy**	**Healthy**	**Healthy**	**Healthy**	**Cancer**
Predicted	Healthy	Healthy	Cancer	Cancer	Cancer
	O	O	X	X	O

Evaluation

- Precision-Recall Traded-off
  - Precision-Recall is a useful measure of success of prediction when the classes are very imbalanced. In information retrieval, precision is a measure of result relevancy, while recall is a measure of how many truly relevant results are returned.
  - Precision-Recall curves summarize the trade-off between the true positive rate and the positive predictive value for a predictive model using different probability thresholds.



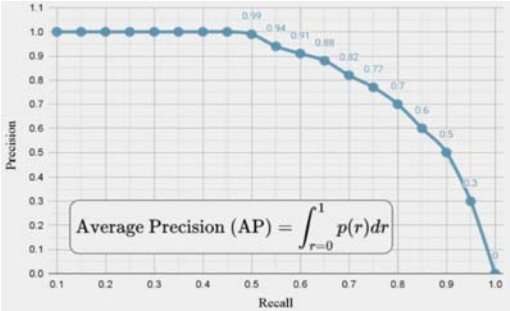
Evaluation

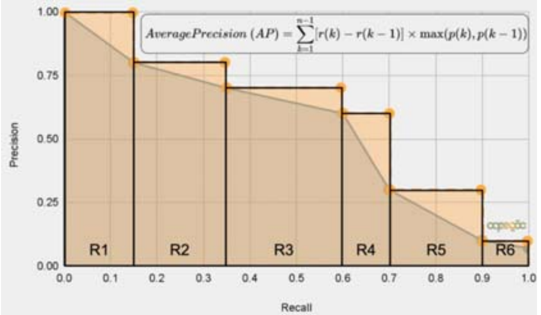
- Average Precision (AP)
  - Average precision is the area under the PR(Precision-Recall) curve
  - AP summarizes the PR Curve to one scalar value. Average precision is high when both precision and recall are high, and low when either of them is low across a range of confidence threshold values. The range for AP is between 0 to 1.

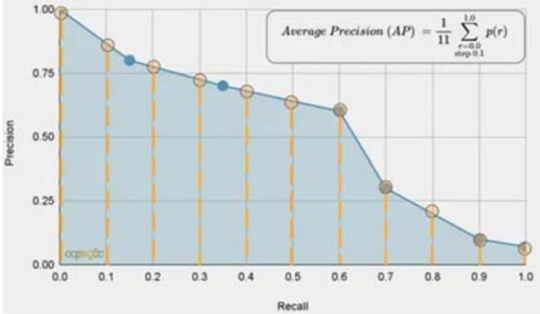
$$Average\ Precision\ (AP) = \int_{r=0}^{1} p(r)dr$$

- Mean Average Precision (mAP)
  - AP value can be calculated for each class. The mean average precision is calculated by taking the average of AP across all the classes under consideration

$$mAP = \frac{1}{k}\sum_{i}^{k} AP_i$$







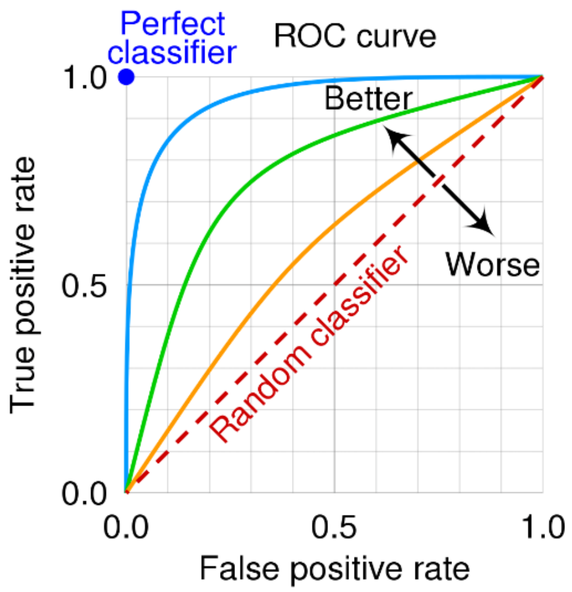
Evaluation

- F1-score
  - It is an alternative machine learning evaluation metric that assesses the predictive skill of a model by elaborating on its class-wise performance rather than an overall performance as done by accuracy.
  - The F1 score combines precision and recall using their harmonic mean, and maximizing the F1 score implies simultaneously maximizing both precision and recall. Thus, the F1 score has become the choice of researchers for evaluating their models in conjunction with accuracy.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

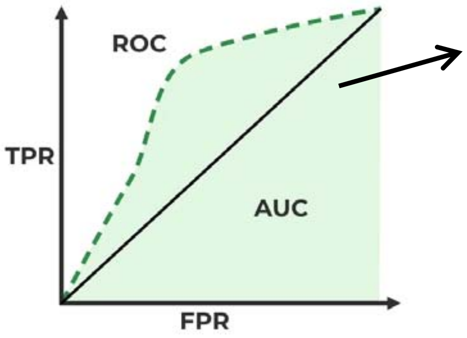
Evaluation

- ROC curve (Receiver Operating Characteristic curve)
  - It is a graph showing the performance of a classification model at all classification thresholds. This curve plots two parameters: True Positive Rate. False Positive Rate.



TPR (True Positive Rate = Recall) : $\frac{TP}{TP+FN}$

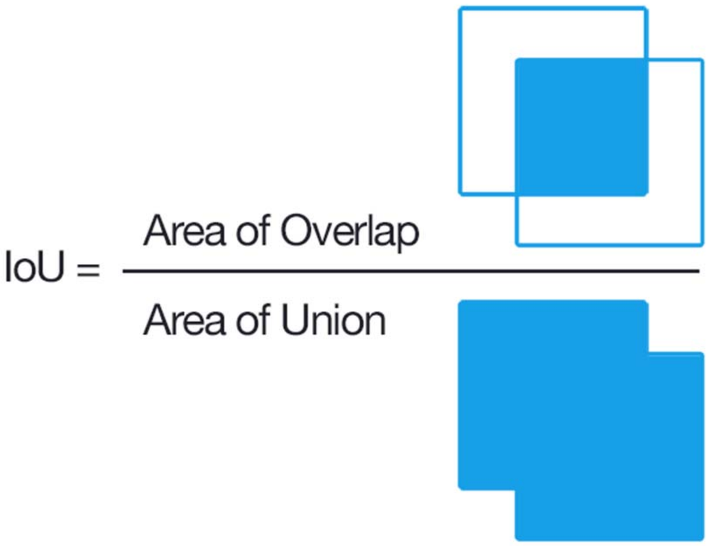
FPR (False Positive Rate = Fall-out) : $\frac{FP}{FP+TN}$



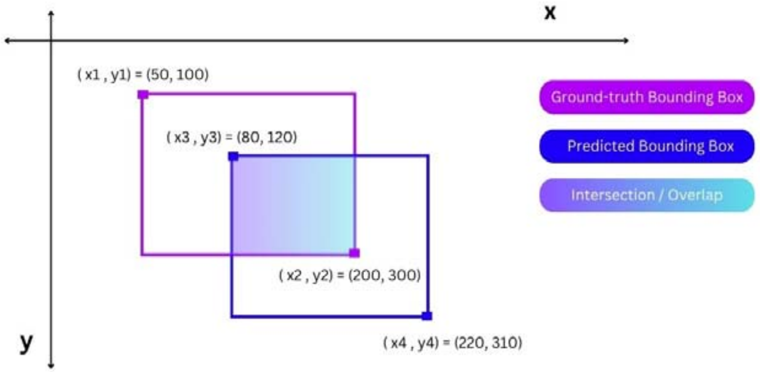
The area under the curve (AUC), as a one-dimensional (uni-dimensional) index, summarizes the "overall" location of the entire ROC curve

Evaluation

- Intersection over Union (IoU)
  - IoU quantifies the closeness of the two bounding boxes (ground truth and prediction). It's a value between 0 and 1. If the two bounding boxes overlap completely, then the prediction is perfect and hence the IoU is 1



$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$


x
y
(x1 , y1) = (50, 100)
(x3 , y3) = (80, 120)
(x2 , y2) = (200, 300)
(x4 , y4) = (220, 310)
Ground-truth Bounding Box
Predicted Bounding Box
Intersection / Overlap

$$Intersection\ over\ Union = \frac{Area\ of\ Intersection}{Area\ of\ Union}$$

$$Intersection\ over\ Union = \frac{21{,}600}{35{,}000}$$

$$Intersection\ over\ Union = 0.618$$

Any Questions?

Thanks!